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The Incident Detection Message Exchange Format version 2 (IDMEFv2)

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Abstract

The Incident Detection Message Exchange Format version 2 (IDMEFv2) defines a data representation for security incidents detected on cyber and/or physical infrastructures.

The format is agnostic so it can be used in standalone or combined cyber (SIEM), physical (PSIM) and availability (NMS) monitoring systems. IDMEFv2 can also be used to represent man made or natural hazards threats.

IDMEFv2 improves situational awareness by facilitating correlation of multiple types of events using the same base format thus enabling efficient detection of complex and combined cyber and physical attacks and incidents.

This draft is maintained by the IDMEFv2 Task Force. Please consult our website for more information: https://www.idmefv2.org.

If approved this draft will obsolete RFC4765.

Status of this Memo

This Internet-Draft is submitted in full conformance with the provisions of BCP 78 and BCP 79.

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1. Introduction

The Incident Detection Message Exchange Format (IDMEF) is intended to solve the problem of security monitoring compartmentalization by proposing a single format to represent any type of incident, whether cyber or physical, intentional or accidental, natural or man-made.

Indeed security is often associated to the Confidentiality-Integrity-Availability triad, performance and availability management systems are still run independently from security management systems.

Additionally, with the adoption and integration of Internet of Things (IoT) and Industrial Internet of Things (IIoT) devices, and the exponential emergence of smart systems (transport, cities, buildings, etc), an increasingly interconnected mesh of cyber-physical systems (CPS) has emerged. This expansion of the attack and incident surfaces blurs the once-clear functions of cybersecurity and physical security.

Finally, as IT infrastructure moves out of data centers it becomes more exposed to external threats, including natural and man-made hazards.

Incident detection systems have traditionally focused on detecting cyber incidents or physical incident or availability incidents. There is an increasing need nowadays to have a unified view and management of all those incidents and their interconnection.

To achieve this goal the Incident Detection Message Exchange Format offers a unique data representation for multiple types of events:

- Cyber-security events (e.g. authentication failure/success, virus/malware detection, bruteforce/scan detection, etc.)
- Physical security events (e.g. intrusion detection, object detection, face or activity recognition, fire/smoke/noise/rain detection, etc.)
- Availability/observability/performance events (e.g. system failure, service malfunction, performance decrease, etc.)
- Natural and man made hazards events (e.g. wildfires, avalanches, droughts, earthquakes, pollution, fire, explosion, etc.)

1.1. IDMEFv2 deployment architecture

IDMEFv2 can be used to exchange incident detection information between specialized managers (SIEM, PSIM, NMS) and a universal "Cyber & Physical SIEM" (CPSIEM) or directly from specialized analyzers and a CPSIEM.

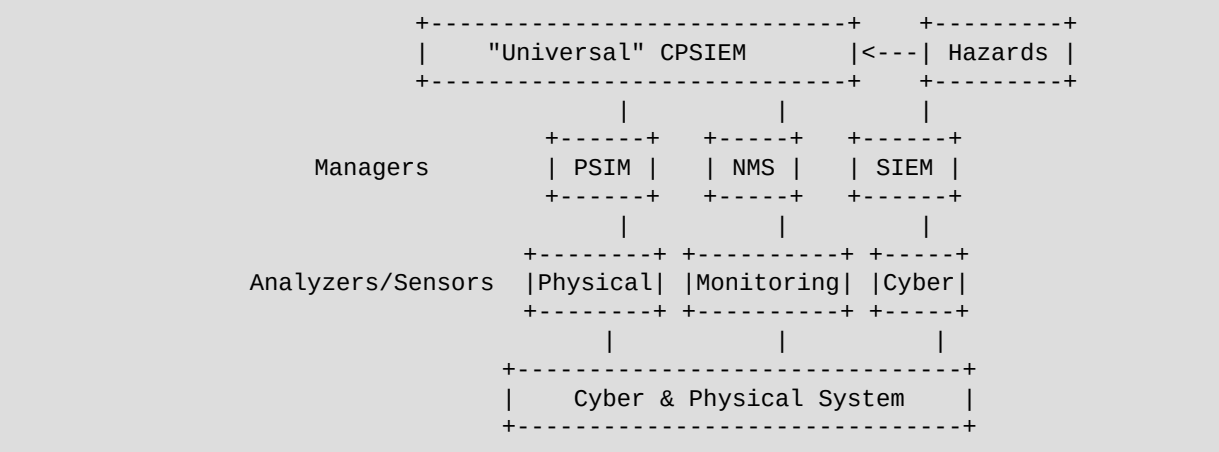


Figure 1: IDMEF Use Architecture

IDMEFv2 improves situational awareness by enabling correlation of multiple types of events using the same base format.

This document defines a model serialization methods for the purpose of describing and sharing these events.

1.2. IDMEFv1 (Intrusion Detection Message Exchange Format) - RFC 4765 - Legacy

IDMEFv2 (Incident Detection Message Exchange Format) is based on IDMEFv1 (Intrusion Detection Message Exchange Format) concepts. But IDMEFv1 was cyber intrusion focused as IDMEFv2 perimeter is much larger. Thus retro-compatibility although partly possible has not been a priority.

1.3. Relationship between IDMEFv2 and other event/incident formats

IDMEFv2 focuses essentialy on high level event/incident correlation and detection.There are many standard and proprietary formats on the incident detection market and in particular on the cybersecurity market. IDMEFv2 is complementary to most of these formats.

IDMEFv1 (Intrusion Detection Message Exchange Format - RFC 4765) : IDMEFv2 (Incident Detection) replaces and obsoletes IDMEFv1 (Intrusion Detection) by covering a wider spectrum.

IODEFv2 (Incident Object Definition Exchange Format - RFC 5070) : IDMEFv2 helps detect incident. When an incident is detected it will be analysed and eventually fully described and shared with other security teams through IODEFv2. IDMEF is used upstream IODEFv2. IDMEFv2 alerts can be “attached” to IODEFv2 object to provide technical details about incidents.

Syslog (System Logging) : Syslog is a lossy format with no formal structure. Syslog can be used by sensors to send information to analyzers. Out of those multi-format syslogs the analyzer might detect an incident or an event of interest. The analyzer will then use IDMEFv2 to notify the manager which might correlate this information with other data to confirm the incident.

SNMP (Simple Network Management Protocol) : SNMP polls information from devices which is then compared to thresholds to detect incident. IDMEFv2 can be used when incident is detected downstream of SNMP to communicate the incident to the manager. IDMEFv2 can have a similar role as SNMP Traps.

STIX (Structured Threat Information Expression) : is a language and serialization format used to exchange cyber threat intelligence (CTI). IDMEFv2 can help detect incidents which might lead to the creation and sharing of STIX information. Cyber analyzer can also rely on STIX information to detect incidents that will be notified in IDMEFv2 format.

OCSF (The Open Cybersecurity Schema Framework) is an open-source, vendor-agnostic standard designed to normalize security telemetry from diverse tools. It provides a common language and consistent structure for security event data, simplifying data ingestion, correlation, and analysis. OCSF can be seen as a “super” syslog to describe events before IDMEFv2 extract “incidents”. OCSF is limited to cyber security.

SIEM proprietory formats (CEF, LEEF, ECS, CIM, ...) : By covering cyber, physical and monitoring incidents type, IDMEFv2 offers a wider spectrum than those formats. Gateways between IDMEFv2 and those formats can be developped to connect legacy cyber detection systems to an IDMEFv2 architecture.

1.4. Existing Deployments and Adoption

IDMEFv2 is not a theoretical proposal. It has been developed, validated, and deployed within the framework of eight large-scale European research projects, funded by the Horizon 2020 and Digital Europe programmes. These projects — namely 7SHIELD, PRECINCT, CyberSEAS, ATLANTIS, ENDURANCE, KINAITICS, TESTUDO, and SAFE4SOC — address critical sectors such as space systems, energy grids, transportation, and government infrastructure. These implementations span multiple domains, including Security Operations Centers (SOCs), Physical Security Information Management (PSIM) systems, and critical infrastructure protection pilots.

This document aims to formalize this existing practice as an IETF Experimental RFC, to ensure interoperability, gather broader community feedback, and provide a stable foundation for future developments.

2. Terminology

2.1. Keywords

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in [RFC2119].

2.2. Normative sections

Implementations of IDMEFv2 are REQUIRED to fully implement:

- The data types defined in Section 3
- The data model defined in Section 5
- The JavaScript Object Notation (JSON) serialization method Section 5.8.

2.3. Concepts related to event processing

2.3.1. Event

An event is something that triggered a notice. Any incident starts off as an event or a combination of events, but not all events result in an incident. An event does not need be an indication of wrongdoing. E.g. someone successfully logging in or entering a building is an event.

2.3.2. Incident

An incident is an event that compromises or has a significant probability of compromising at least one of the organization's security criteria such as Confidentiality, Integrity or Availability. An incident may affect a production tool, personnel, etc. It may be logical, physical or organizational in nature. Last but not least, an incident may be caused on purpose or by accident.

2.3.3. Alert

An alert is a notification/message that a particular event/incident (or series of events/incidents) has occurred.

2.3.4. Manager

The manager is the central console toward which all analyzers send their alerts. The manager collects, correlates, stores and displays the alerts to the operators.

Example : - A SIEM (Security Information & Event Management) or a Log Manager) - A PSIM (Physical Security Information Management) - A NMS (Network Management System) - A CPSIEM (Cyber & Physical Security Information Management System)

2.3.5. Operator

The level 1 operator is in charge of receiving manager notifications and identify or confirm when an event should be considered as an incident. The operator must also decide if there is a known resolution for this incident or if it needs a deeper analysys.

2.3.6. Analyst

The analyst will be contacted by the operator to analyze complex incidents that can’t be easily resolved. The investigation starts with the IDMEFv2 information but the analyst might need more information like raw logs for a deeper forensics.

2.3.7. Attack

An attack is an attempt to destroy, expose, alter, disable, steal or gain unauthorized access to or make unauthorized use of a cyber or physical asset. An attack is one or many kinds of incidents.

2.3.8. Correlation

Correlation is the identification of relationships between two or more events.

2.3.9. Aggregation

Aggregation is the consolidation of similar events into a single event.

3. The IDMEF Data Types

Each object inside the IDMEF data model has an associated data type. This type may be used to validate the content of incoming IDMEF messages.

3.1. Classes

The classes are meant to group related attributes together. Some of the classes may be instanciated multiple times (e.g. Source, Target, etc.) while others may only appear once in an IDMEF message (e.g. Analyzer).

3.2. Numbers

3.2.1. Integers

Integers inside the IDMEF data model are expressed using the following ABNF [RFC5234] grammar:

```
integer         =  *1minus int
int             =  zero / ( digit1-9 *DIGIT )
minus           =  %x2D                          ; -
zero            =  %0x30                         ; 0
digit1-9        =  %x31-39                       ; 1-9
```

E.g. 123.

Such values are indicated with the "INT" type annotation in the model.

3.2.2. Floating-point values

Floating-point values inside the IDMEF data model are expressed using the following ABNF grammar:

```
float           =  integer *1frac
frac            =  decimal-point 1*DIGIT
decimal-point   =  %x2E                           ; .
```

This grammar reuses some of the production rules listed in Section 3.2.1.

E.g. 12.34.

Such values are indicated with the "FLOAT" type annotation in the model.

3.3. Strings

Strings are series of characters from the [UNICODE] standard and are used to represent a text.

For readability, this document uses quotes (") to delimit strings, but please note that these quotes are not syntactically part of the actual strings.

E.g. "Hello world".

Some of the strings used in the IDMEFv2 data model follow a stricter syntax. These are included below for completeness.

Such values are indicated with the "STRING" type annotation in the model.

3.3.1. Enumerations

Enumerations are special strings used when valid values for an IDMEF attribute are restricted to those present in a predefined list.

Such values are indicated with the "ENUM" type annotation in the model.

3.3.2. Timestamps

Timestamps are used to indicate a specific moment in time. The timestamps used in the IDMEF data model follow the syntax defined by the "date-time" production rule of the grammar in [RFC3339] ch 5.6.

E.g. "1985-04-12T23:59:59.52Z" represents a moment just before April 5th, 1985 in Coordinated Universal Time (UTC).

Such values are indicated with the "TIMESTAMP" type annotation in the model.

3.3.3. Geographical Locations

Some attributes inside the IDMEF data model may refer to geographical locations using a set of coordinates. The reference system for all geographical coordinates is a geographic coordinate reference system, using the World Geodetic System 1984 [WGS84]. The reference system used is the same as for the Global Positioning System (GPS).

The format for such values can be either "latitude,longitude" or "latitude,longitude,altitude". Each of these coordinates is represented as a floating-point value. The latitude and longitude are expressed in degrees while the altitude is expressed in meters.

E.g. "48.8584,2.2945,276.13" matches the (3-dimensional) geographical location for the top floor or the Eiffel Tower located in Paris, France, while "48.8584,2.2945" matches the same location in two dimensions (with the altitude removed).

Such values are indicated with the "GEOLOC" type annotation in the model.

3.3.4. UNECE Location Codes (UN/LOCODE)

Some attributes inside the IDMEF data model may refer to geographical locations using Locations Codes. These codes can be assimilated to an enumeration, where the list of possible values is defined in the United Nations Economic Commission for Europe (UNECE) Codes for Trade [UN-LOCODE].

E.g. "FR PAR" is the Location Code for the city of Paris, France.

Such values are indicated with the "UNLOCODE" type annotation in the model.

3.3.5. Uniform Resource Identifiers (URIs)

The IDMEF data model uses Uniform Resource Identifiers (URIs), as defined in [RFC3986], when referring to external resources. Unless otherwise specified, either a Uniform Resource Location (URL) or a Uniform Resource Name (URN) may be used where a URI is expected.

E.g. both "https://example.com/resource" and "urn:myapp:resource" are valid Uniform Resource Identifiers.

Such values are indicated with the "URI" type annotation in the model.

3.3.6. IP Addresses

IP addresses inside the IDMEF data model are expressed as strings using the traditional dotted-decimal notation for IPv4 addresses (defined by the "dotnum" production rule in the grammar in [RFC5321]), while IPv6 addresses are expressed using the text representation defined in [RFC4291] ch 2.2.

E.g. "192.0.2.1" represents a valid IPv4 address, while "::1/128" represents a valid IPv6 address.

It is RECOMMENDED that implementations follow the recommendations for IPv6 text representation stated in [RFC5952].

Such values are indicated with the "IP" type annotation in the model.

3.3.7. E-mail addresses

E-mail addresses inside the IDMEF data model are expressed as strings using the address specification syntax defined in [RFC5322] ch 3.4.1.

E.g. "root@example.com".

Such values are indicated with the "EMAIL" type annotation in the model.

3.3.8. Attachment names

Attachments inside the IDMEF data model are identified using a unique name, composed of a string whose character set is limited to the ASCII letters (A-Z a-z) and digits (0-9).

E.g. "state" is a valid name for an attachment.

The constraint on name unicity is enforced per class. That is, but it is not possible for two attachments to share the same name inside the same alert.

Such values are indicated with the "ID" type annotation in the model.

3.3.9. Media types

Media types are used in the IDMEF data model to describe an attachment's content. The syntax for such values is defined in [RFC2046].

IANA keeps a list of all currently registered media types in the Media Types registry .

E.g. "application/xml" or "text/plain; charset=utf-8".

Such values are indicated with the "MEDIATYPE" type annotation in the model.

3.3.10. Universally Unique IDentifiers (UUIDs)

Universally Unique Identifiers (UUIDs) are used to uniquely identify IDMEF messages. It is also possible for an IDMEF message to reference other IDMEF messages using their UUIDs. The syntax for UUIDs is defined in [RFC4122].

To limit the risk of UUID collisions, implementors SHOULD NOT generate version 4 UUIDs (randomly or pseudo-randomly generated UUIDs).

E.g. "ba2e4ef4-8719-42bb-a712-d6e8871c5c5a".

UUIDs are case-insensitive when used in comparisons.

Such values are indicated with the "UUID" type annotation in the model.

3.3.11. Protocol Names

Such values are indicated with the "PROTOCOL" type annotation in the model.

3.3.12. IDMEF Paths

This document defines a way to represent the path to every possible attribute inside an IDMEF message. For conciseness, the top-level "Alert" class is omitted from the path.

This representation can be used in contexts where the path to an IDMEF attribute is expected. An example of such usage can be seen in the definition of the "AggrCondition" attribute inside the Alert class (Section 5.2).

The syntax for these IDMEF paths is expressed in the following ABNF grammar:

```
class-name      =  "Analyzer" / "Sensor" / "Source" / "Target" /
                   "Attachment"
attribute-name  =  1*ALPHA
class-reference =  class-name "."
num             =  *1"-" 1*DIGIT
list-index      =  "(" num ")"
path            =  *1class-reference attribute-name *1list-index
```

Valid attribute names are limited to those defined for the specified class-reference (or in the top-level "Alert" class if class-reference is omitted).

For example, the following path refers to the "CeaseTime" attribute of the top-level "Alert" class: "CeaseTime".

Likewise, the following path refers to the "Name" attribute of the "Analyzer" class: "Analyzer.Name".

For attributes defined as lists (see Section 3.4), the path may include the (0-based) index for an entry inside the list. The index defaults to 0 if omitted. This means that several (valid) representations may be used to reference the same IDMEF attribute when list attributes are involved.

For example, both of the following paths refer to the IP address of the first source associated with an IDMEF message:

```
Source.IP
Source(0).IP
```

Compatible implementations MUST reject paths that reference an unknown class, an unknown attribute, or use a list-index for an IDMEF field which is not defined as a list.

A compatible implementation MUST also normalize paths before comparing them (e.g. by stripping the text "(0)" from paths referring to list attributes).

3.3.13. Hashes

Hashes are sometimes used inside the data model to protect the integrity (and optionally, authenticity) of attachments.

The syntax for these values is "function:hash_result", where "function" refers to one of the hashing function names listed in and "hash_result" contains the hexadecimal notation for the hash result obtained by calling the specified hash function on the input value.

In the context of IDMEF, either a keyless or keyed hash function may be used to process the raw input value.

E.g. "sha256:a02735ed8b10ad432d557bd4849c0dac3b23d64706e0618716d6df2def338374"

Hashes are case-insensitive when used in comparisons.

Such values are indicated with the "HASH" type annotation in the model.

3.4. Lists

Some attributes of the IDMEF data model accept ordered lists of values.

Such ordered lists are indicated with the "X[]" type annotation in the model. where "X" refers to one of the data types defined in Section 3. For example, "ENUM[]" refers to an ordered list of enumeration values.

4. The IDMEF extension

In order to support the dynamic nature of security operations and to adapt to specific needs, the IDMEFv2 data model will need to continue to evolve. This section discusses how new data elements can be incorporated into the IDMEFv2. There is support to add additional enumerated values and new attributes.

These extension mechanisms are designed so that adding new data elements is possible without requiring modifications to this document. Extensions can be implemented publicly or privately. With proven value, well-documented extensions can be incorporated into future versions of the specification.

4.1. Extending the Enumerated Values of Attributes

Additional enumerated values can be added to select attributes either through the use of specially marked attributes with the "ext-" prefix or through a set of corresponding IANA registries. The former approach allows for the extension to remain private. The latter approach is public.

4.1.1. Private Extension of Enumerated Values

The data model supports adding new enumerated values to an attribute without public registration. For each attribute that supports this extension technique, there is a corresponding attribute in the same element whose name is identical but with a prefix of "ext-". This special attribute is referred to as the extension attribute. The attribute being extended is referred to as an extensible attribute. For example, an extensible attribute named "foo" will have a corresponding extension attribute named "ext-foo". An element may have many extensible attributes.

In addition to a corresponding extension attribute, each extensible attribute has "ext-value" as one its possible enumerated values. Selection of this particular value in an extensible attribute signals that the extension attribute contains data. Otherwise, this "ext-value" value has no meaning.

In order to add a new enumerated value to an extensible attribute, the value of this attribute MUST be set to "ext-value", and the new desired value MUST be set in the corresponding extension attribute. For example, extending the Category attribute of the Analyzer class would look as follows:

```
Analyzer: {
    ...
    "Category":["ext-value"],
    "ext-Category": "my-new-analyzer-category",
    ....
}
```

A given extension attribute MUST NOT be set unless the corresponding extensible attribute has been set to "ext-value".

4.1.2. Public Extension of Enumerated Values

The data model also supports publicly extending select enumerated attributes. A new entry can be added by registering a new entry in the appropriate IANA registry. Section (Table 5) provides a mapping between the extensible attributes and their corresponding registry.

4.2. Private Extension of Attributes

Use of new attributes is possible through the use of the attachment class. New attributes and their corresponding values should be stored in the Content attribute of an Attachment and the ContentEncoding must be set to JSON. For example creating a new attribute to store the email of the operator (in charge of solving the incident) will look as follows:

```
"Attachment" : [
    {
    "Name": "Operator",
    "ContentEncoding": "JSON",
    "Content": "{\"OperatorMail\":\"John.Does@acme.com\"}",
    }
[
```

5. The IDMEF Data Model

In this section, the individual components of the IDMEF data model will be discussed in detail. For each class, the semantics will be described.

5.1. Overview

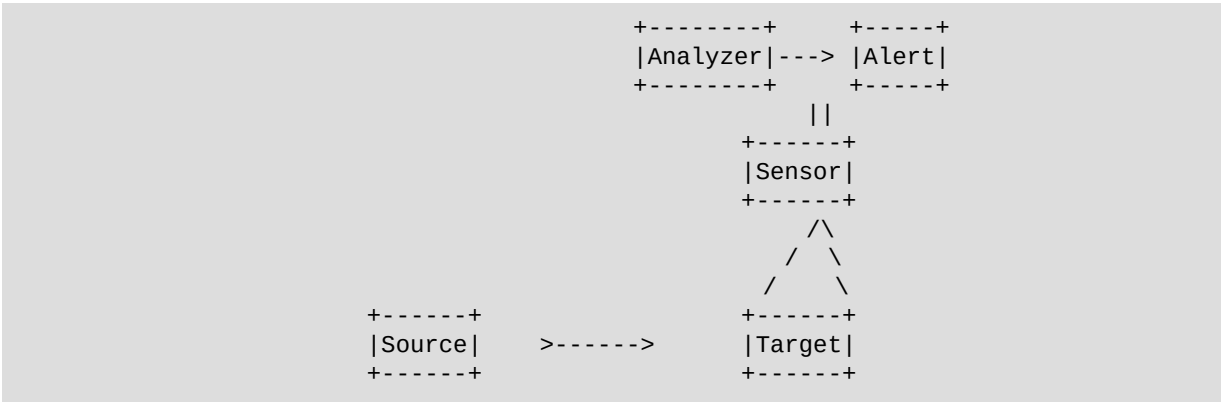


Figure 2: IDMEFv2 Overview Classes

An IDMEF message is composed of an instance of the Alert class (Section 5.2) representing the overall properties of the message. It also contains exactly one instance of the Analyzer class (Section 5.3) and zero or more instances of the Sensor class (Section 5.4). The message may also describe various aspects of an event using the Source (Section 5.5) and Target (Section 5.6) classes.

Last but not least, it may also include zero or more instances of the Attachment class (Section 5.7), e.g. captured files or network packets related to the event for example.

The relationship between the main Alert class and other classes of the data model is shown in Figure 3 (attributes are omitted for clarity).

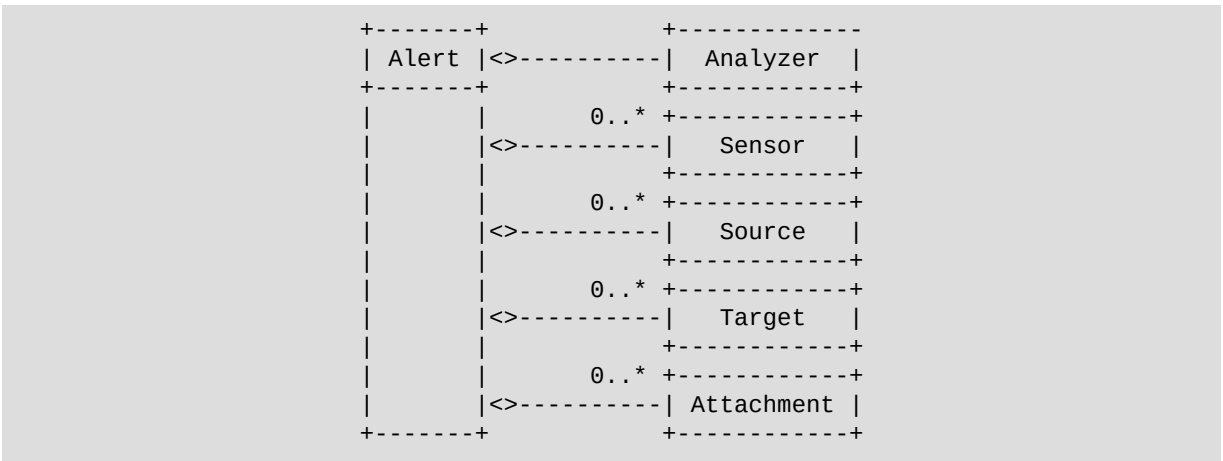


Figure 3: IDMEFv2 Classes

It is important to note that the data model does not specify how an incident should be categorized or identified. For example, an attacker scanning a network for machines listening on a specific port may be identified by one analyzer as a single attack against multiple targets, while another analyzer may identify it as multiple attacks from a single source. However, once an analyzer has determined the type of alert it plans on sending, the data model dictates how that alert should be formatted.

5.2. The Alert Class

The Alert class contains high level information about the event that triggered the alert.



Figure 4: The Alert class

The aggregate classes that make up Alert are:

Analyzer	Exactly one. An instance of the Analyzer class (Section 5.3) that describes the tool/device responsible for the analysis that resulted in the alert being created and sent.
Sensor	Zero or more. Instances of the Sensor class (Section 5.4) used to describe the sensor(s) that captured the information used during the analysis. Depending on the tools/devices used to detect incidents, an Analyzer may rely on the output from a single sensor or from multiple sensors to generate alerts. In addition, the Analyzer and Sensor may actually be part of the same physical device and may share some of their attributes (e.g. IP, Hostname, Model, etc.).
Source	Zero or more. Instances of the Source class (Section 5.5) used to describe the source(s) of the incident (e.g. attackers, faulty device, etc.).
Target	Zero or more. Instances of the Target class (Section 5.6) used to describe the target(s) of the incident, i.e. the impacted devices/users/services/locations.

Attachment	Zero or more. Instances of the Attachment class (Section 5.7) used to describe the electronic artifacts captured in relation with the event. The intent of the Attachment class is to keep track of the electronic files left as a trail during the event. This may include things like on-disk files (e.g. malware samples), network packet captures, videos or still images from a camera feed, voice recording, etc.

The Alert class has the following attributes:

Version	Mandatory. The version of the IDMEF format in use by this alert. During the drafts tuning period the version is equal to the draft version. Therefore it is "2.D.V0X" for Draft V0X.
ID	Mandatory. Unique identifier for the alert.
OrganisationName	Optional. Corporate/Main Office Organisation Name Useful if alerts are sent to a multi-organisation central incident detection manager. Example: ACME Corporation
OrganisationId	Optional. Corporate/Main Office Organisation ID. Where possible official organisation ID manage by national authority. Useful if alerts are sent to a multi-organisation central incident detection manager. This ID has to be chosen depending on the overall detection perimeter and the nature of the monitored organisation (Private/Public, Commercial, International, etc.) Examples: OrganisationId in France could be SIREN, in England could be CR, Germany could be Handelsregisternummer, Spain could be CIF, Italia could be Partita IVA, USA could be EIN, etc. Commercial OrganisationId in Europe could be V.A.T ID
EntityName	Optional. Entity Name, monitored by the organisation, where the incident occurred. Could be a town, region or country name or an internal name. Could also be the name of a customer for a MSSP centralizing it's customer incidents in a single system. Do not repeat the organisation name in the EntityName Example: - ACME HeadQuaters is located in Paris France and has a local office in India - If the incident occurred in the local office: "OrganisationName": "ACME" , "EntityName": "India" - If the incident occurred in the headquaters: "OrganisationName": "ACME", "EntityName": "Headquaters" (or "Paris")
EntityId	Optional. Entity ID, monitored by the organisation, where the incident occurred. Useful if organisation and entity are not directly linked, like a customer and a MSSP.
EntitySector	Optional. The economic sector(s) and sub-sector(s) in which the entity operates. Values follow the dot notation sector.subsector based on the critical infrastructure taxonomy defined in the NIS2 Directive and CER (Critical Entities Resilience) Directive. This attribute enables sector-based correlation, regulatory compliance reporting, and risk context for incident detection. Enumeration available here Table 6

Type — Optional. Incident type.

Rank	Keyword	Description
0	Cyber	Cyber incident
1	Physical	Physical incident
2	Availability	Availability incident
3	NatHaz	Natural Hazards incident
4	Combined	Combined incident

Table 1: Incident types

Category — Optional. Incident category.
Enumeration available here Table 7

ext-Category — Optional. A means by which to extend the Category attribute. (see Section 4.1.1)

Cause — Optional. Incident cause. The cause can be modified by any analyzer on the way of the alert and later by the operator and/or the analyst if new investigation reveals and confirms a different cause of the event.

Rank	Keyword	Description
0	Normal	The event is related to an expected phenomenon or to a phenomenon that does not qualify as out of the ordinary.
1	Error	The event is related to a human error.
2	Malicious	The event is related to malicious code or malicious actions.
3	Malfunction	The event is related to a device or service malfunction.
4	Hazard	The event is related to a hazard phenomenon.
5	Unknown	The cause of the event is unknown.

Table 2: Incident causes

Description — Optional. Short free text human-readable description of the event.
The description can add detail to the event category for easiest/faster comprehension by the operator. Example : * Cryptoware WannaCry blocked on pegasus server * Unknown person entering through east doorway

Status — Optional. Event state in the overall event lifecycle.

Rank	Keyword	Description
0	Event	The event is still considered as an harmless event and should not be treated.
1	Incident	The event is considered as an incident and should be taken care of.
2	Open	The incident is confirmed and actively being investigated.
3	Closed	Investigation is finished and the issue is handled.

Rank	Keyword	Description
4	FalsePositive	Investigation is finished, incident was a false positive.
5	Obsolete	Incident is obsolete, usually by another incident.

Table 3: Incident statuses

Priority — Optional. Priority of the alert. Priority is defined by combining impact and urgency. It indicates how fast the incident should be taken care of.

Rank	Keyword	Description
0	Unknown	Priority unknown
1	Info	Informational event with no current impact; no action or follow-up is required.
2	Low	Minor incident with negligible impact; a workaround exists, and resolution can be scheduled.
3	Medium	Moderate incident affecting a limited scope; requires timely attention but is not immediately business-threatening.
4	High	Significant incident with major service degradation or impact on important assets; demands an urgent response.
5	Critical	Severe incident with business-critical systems down or vital interests threatened; requires immediate action and escalation.

Table 4: Incident priorities

Confidence — Optional. A floating-point value between 0 and 1 indicating the analyzer's confidence in its own reliability of this particular detection, where 0 means that the detection is surely incorrect while 1 means there is no doubt about the detection made.

Note — Optional. Free text human-readable additional note, possibly a longer description of the incident if is not already obvious.
The Note attribute can be used to store any additional information. It can be additional information about the event and/or about the incident resolution, although the incident resolution information should in principle be stored elsewhere (with a link with the external tool in AltNames)

CreateTime — Mandatory. Timestamp indicating when the alert was created.

StartTime — Optional. Timestamp indicating the deduced start of the event.
StartTime can be later than CreateTime in case or Alerts created from forecast information (e.g. Snow Storm expected in two days staring at 10h00)

EndTime — Optional. Timestamp indicating the deduced end of the event.

ReportTime — Optional. Timestamp indicating the reporting time of the event, usually to an external CSIRT or a central SIEM.

AltNames	Optional. Alternative identifiers; strings which help pair the event to internal systems' information (for example ticket IDs inside a request tracking systems).
AltCategory	Optional. Alternate categories from a reference other than IDMEFv2 categories (e.g. MISP, MITRE ATT@CK or another proprietary/internal reference).
Ref	Optional. References to sources of information related to the incident and/or vulnerability, and specific to this incident. This MAY be a URL to additional info, or a URN in a registered or unregistered ad-hoc namespace bearing reasonable information value and uniqueness, such as "urn:cve:CVE-2013-2266".
CorrelID	Optional. Identifiers for the messages which were used as information sources to create this message, in case the message has been created based on correlation/analysis/deduction from other messages.
AggrCondition	Optional. A list of IDMEF fields used to aggregate events. The values for these fields will be the same in all aggregated events. This attribute should mostly be set by intermediary nodes, which detect duplicates, or aggregate events, spanning multiple detection windows, into a longer one. The "StartTime" and "EndTime" attributes are used in conjunction with this attribute to describe the aggregation window.
PredID	Optional. A list containing the identifiers of previous messages which are obsoleted by this message. The obsoleted alerts SHOULD NOT be used anymore. This field can be used to "update" an alert.
RelID	Optional. A list containing the identifiers of other messages related to this message.

5.3. The Analyzer Class

The Analyzer class describes the module that has analyzed the data captured by the sensors, identified an event of interest and decided to create an alert.

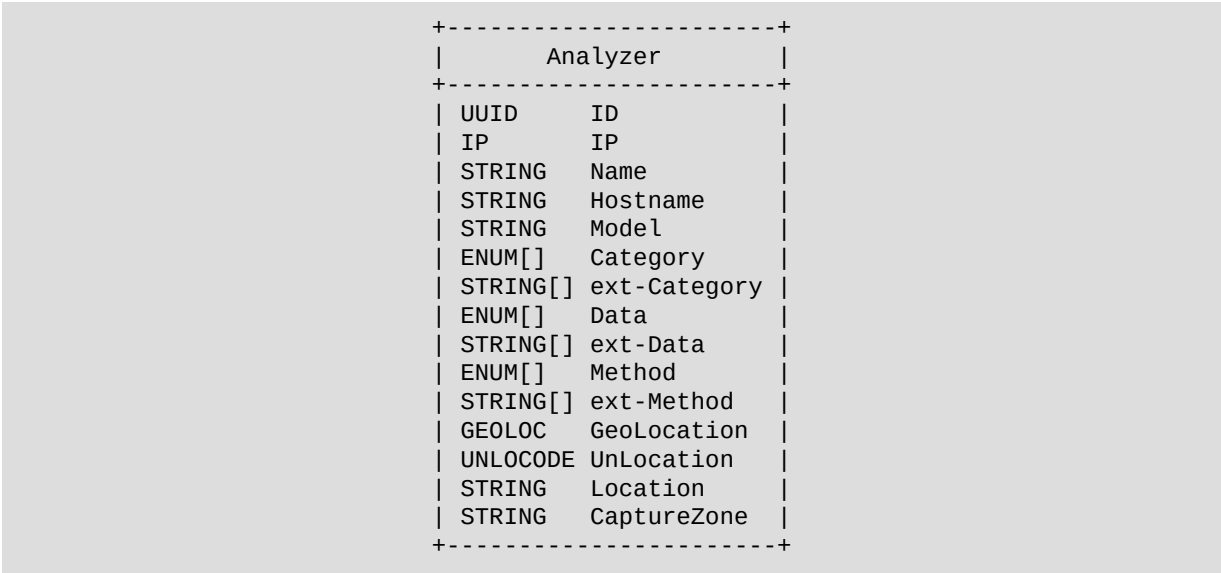


Figure 5: The Analyzer class

The Analyzer class has the following attributes:

ID	Optional. Unique identifier for the analyzer.
IP	Optional. Analyzer IP address.
Name	Recommended. Name of the analyzer, which must be reasonably unique, however still bear some meaningful sense. This attribute usually denotes the hierarchy of organizational units the detector belongs to and its own name. It MAY also be used to distinguish multiple analyzers running with the same IP address.
Hostname	Optional. Hostname of this analyzer. SHOULD be a fully-qualified domain name.
Model	Optional. Analyzer model description (usually its generic name, brand and version).
Category	Optional. Analyzer categories. Enumeration available here Table 8
ext-Category	Optional. A means by which to extend the Category attribute. (see Section 4.1.1)
Data	Optional. Type of data analyzed during the detection. Enumeration available here Table 9
ext-Data	Optional. A means by which to extend the Data attribute. (see Section 4.1.1)
Method	Optional. Detection method. Enumeration available here Table 10
ext-Method	Optional. A means by which to extend the Method attribute. (see Section 4.1.1)
GeoLocation	Optional. GPS coordinates for the analyzer.
UnLocation	Optional. Standard UN/Locode for the analyzer.
Location	Optional. Internal name for the location of the analyzer.
CaptureZone	Optional. A string that describes the "capture zone" of the analyzer (if the analyzer is also a sensor), as a JSON-serialized string. Depending on the type of sensor, the capture zone may for instance refer to: A JSON object describing a camera's settings (elevation, horizontal and vertical field of view, azimuth, etc.) A description of the IP network where packet capture is taking place.

5.4. The Sensor Class

The Sensor class describes the module that captured the data before sending it to an analyzer. The Sensor may be a subpart of the Analyzer.

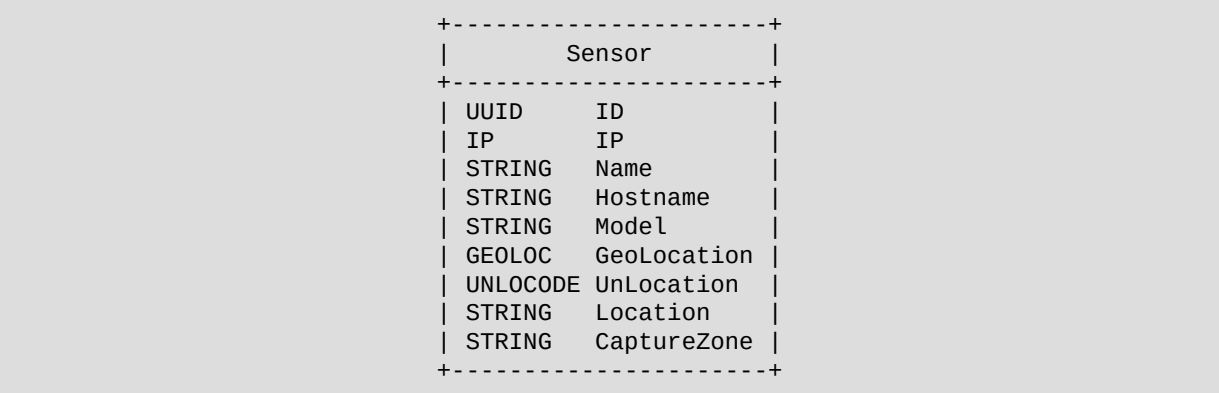


Figure 6: The Sensor class

The Sensor class has the following attributes:

ID	Optional. Unique identifier for the sensor.
IP	Optional. The sensor's IP address.
Name	Recommended. Name of the sensor, which must be reasonably unique, however still bear some meaningful sense. This attribute usually denotes the hierarchy of organizational units the sensor belongs to and its own name. It MAY also be used to distinguish multiple sensors running with the same IP address.
Hostname	Optional. The sensor's hostname. This SHOULD be a fully qualified domain name, but may not conform exactly because values extracted from logs, messages, DNS, etc. may themselves be malformed. An empty string MAY be used to explicitly state that this value was inquired but not found (missing DNS entry).
Model	Optional. The sensor model's description (usually its generic name, brand and version).
GeoLocation	Optional. GPS coordinates for the analyzer.
UnLocation	Optional. Standard UN/Locode for the sensor.
Location	Optional. Internal name for the location of the sensor.
CaptureZone	Optional. A string that describes the "capture zone" of the sensor, as a JSON-serialized string. Depending on the type of sensor, the capture zone may for instance refer to:

- A JSON object describing a camera's settings (elevation, horizontal and vertical field of view, azimuth, etc.)
- A description of the IP network where packet capture is taking place.

5.5. The Source Class

The Source class describes the origin(s) of the event(s) leading up to the creation of this alert.

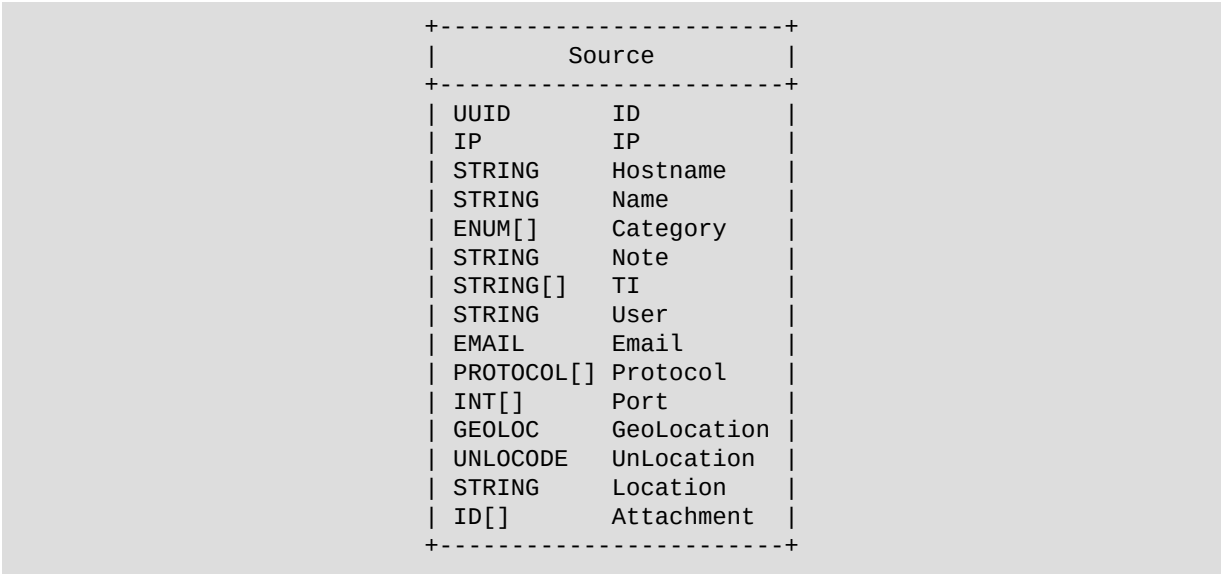


Figure 7: The Source class

The Source class has the following attributes:

ID	Optional. Unique identifier for the source.

IP	Optional. Source IP address.
Hostname	Optional. Hostname of this source. This SHOULD be a fully qualified domain name, but may not conform exactly because values extracted from logs, messages, DNS, etc. may themselves be malformed. An empty string MAY be used to explicitly state that this value was inquired but not found (missing DNS entry).
Name	Optional. Name of this source. Naming the source might be necessary for non digital source like a human, a vehicle, a drone, etc.
Category	Optional. Source category Enumeration available here Table 11
Note	Optional. Free text human-readable additional note for this source.
TI	Optional. Threat Intelligence data about the source. Values in this list MUST use the format "attribute:origin", where "attribute" refers to the attribute inside this source found inside a Threat Intelligence database, and "origin" contains a short identifier for the Threat Intelligence database. E.g. "IP:Dshield". Please note that the same attribute may appear multiple times inside the list (because a match was found in multiple Threat Intelligence databases).
User	Optional. User ID or login responsible for the incident.
Email	Optional. Email address responsible for the incident. E.g. the value of the "Reply-To" or "From" header inside a phishing e-mail.
Protocol	Optional. Protocols related to connections from/to this source. If several protocols are stacked, they MUST be ordered from the lowest (the closest to the medium) to the highest (the closest to the application) according to the ISO/OSI model.
Port	Optional. Source ports involved in the incident. Values in this list MUST be integers and MUST be in the range 1-65535.
GeoLocation	Optional. GPS coordinates for the source.
UnLocation	Optional. Standard UN/Locode for the source.
Location	Optional. Internal name for the location of the source.
Attachment	Optional. Identifiers for attachments related to this source. Each identifier listed here MUST match the "Name" attribute for one of the attachments described using the Attachment class (Section 5.7).

5.6. The Target Class

The Target class describes the target(s) impacted by the event(s) leading up to the creation of this alert.

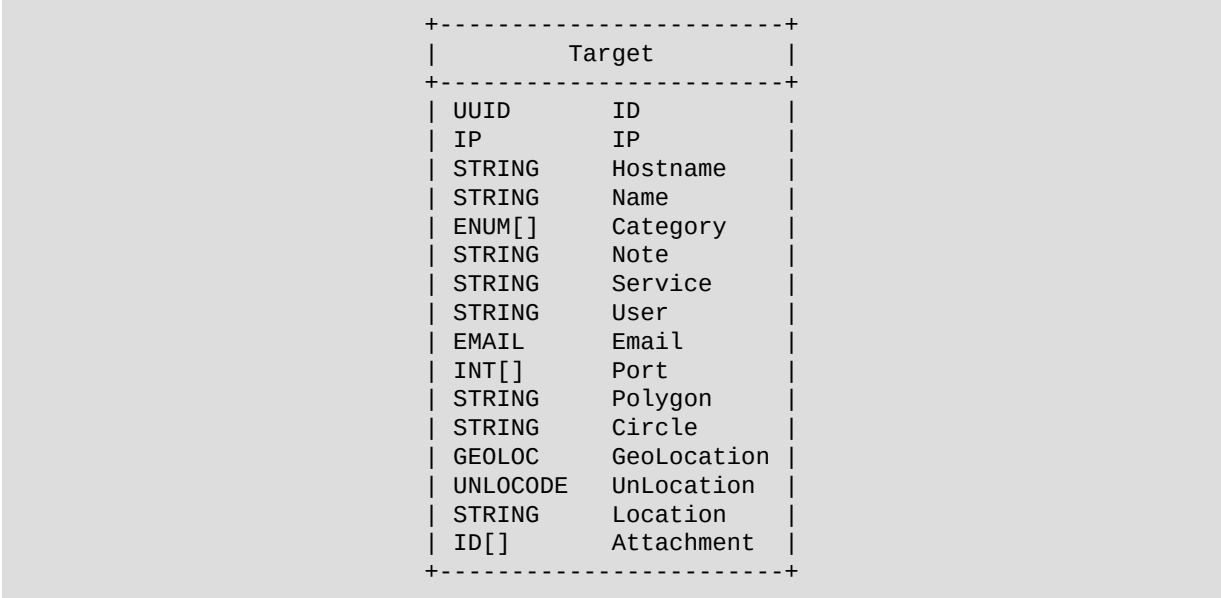


Figure 8: The Target class

The Target class has the following attributes:

ID	Optional.Unique identifier for the target.
IP	Optional. Target IP address.
Hostname	Optional. Hostname of this target. This SHOULD be a fully qualified domain name, but may not conform exactly because values extracted from logs, messages, DNS, etc. may themselves be malformed. An empty string MAY be used to explicitly state that this value was inquired but not found (missing DNS entry).
Name	Optional. Name of this target. Naming the target might be necessary for non digital targets like a human, a vehicle, a building, etc.
Category	Optional. Target category Enumeration available here Table 12
Note	Optional. Free text human-readable additional note for this target.
Service	Optional. Should be High Level Service (messaging, internet, ...), if not defined can be process impacted by the incident.
User	Optional. User ID or login impacted by the incident.
Email	Optional. Email address impacted by the incident. E.g. the value of the "To" header inside a phishing e-mail.
Port	Optional. Target ports involved in the incident. Values in this list MUST be integers and MUST be in the range 1-65535.
Polygon	Optional. Pollygonal incident affected area. Whitespace-delimited list of WGS 84 'latitude,longitude' pairs. Minimum 4 pairs, first equals last
Circle	Optional. Circular incident infected area. WGS 84 center point 'latitude,longitude' followed by a space and radius in kilometers
GeoLocation	Optional. GPS coordinates for the target.
UnLocation	Optional. Standard UN/Locode for the target.

Location	Optional. Internal name for the location of the target.
Attachment	Optional. Identifiers for attachments related to this target. Each identifier listed here MUST match the "Name" attribute for one of the attachments described using the Attachment class (Section 5.7).

5.7. The Attachment Class

The Attachment class contains additional data which was captured in relation with the event.

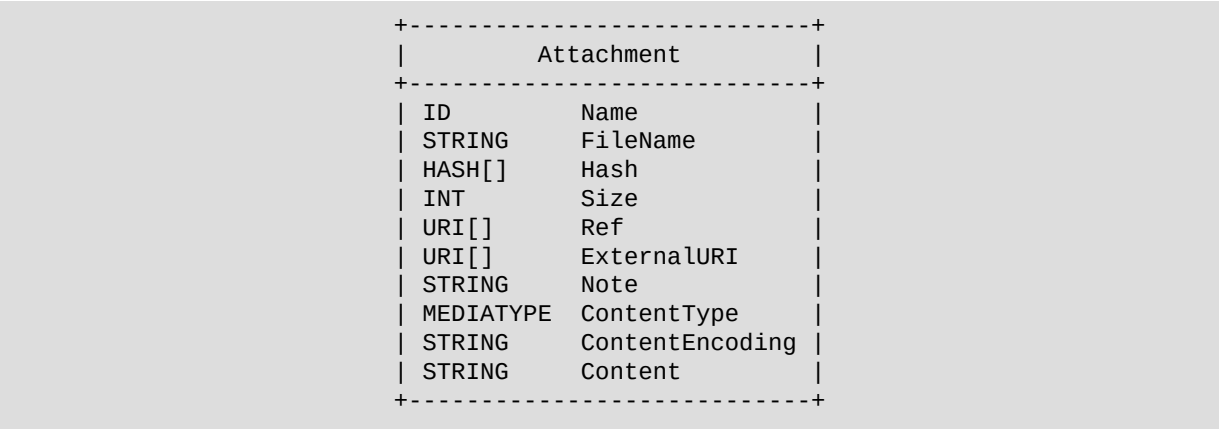


Figure 9: The Attachment class

The Attachment class has the following attributes:

Name	Mandatory. A unique identifier among attachments that can be used to reference this attachment from other classes using the "Attachment" attribute.
FileName	Optional. Attachment filename. This will usually be the original name of the captured file or the name of the file containing the captured content (e.g. a packet capture file).
Hash	Optional. A list of hash results for the attachment's Content. The values in this list are computed by taking the raw value of the attachment's "Content" attribute. The hash result is computed before any other transformation (e.g. Base64 encoding) is applied to the content, so that a receiving IDMEF system may reverse the transformation, apply the same hashing function and obtain the same hash result. See also the definition for the "ContentEncoding" attribute below. It is RECOMMENDED that compatible implementations use one of the hashing functions from the SHA-2 [RFC6234] or SHA-3 [NIST.FIPS.202] families to compute the hash results in this list.
Size	Optional. Length of the content (in bytes). This value MUST be a non-negative integer.
Ref	Optional. References to sources of information related to the incident and/or vulnerability, and specific to this attachment.
ExternalURI	Optional. If the attachment's content is available and/or recognizable from an external resource, this is the URI (usually a URL) to that resource. This MAY also be a URN in a registered or unregistered ad-hoc namespace bearing reasonable information value and uniqueness, such as "urn:mhr:55eaf7effadc07f866d1eaed9c64e7ee49fe081a" or "magnet:?xt=urn:sha1:YNCKHTQCWBTRNJIV4WNAE52SJUQCZO5C".

Note	Optional. Free text human-readable additional note for this attachment.
ContentType	Optional. Internet Media Type of the attachment. For compatibility reasons, implementations SHOULD prefer one of the well-known media types registered in IANA .
ContentEncoding	Optional. Content encoding. The following encodings are defined in this version of the specification: "json": The content refers to a JSON object which has been serialized to a string using the serialization procedure defined in [RFC8259]. "base64": The content has been serialized using the Base64 encoding defined in [RFC4648]. The "base64" encoding SHOULD be used when the content contains binary data. If omitted, the "json" encoding MUST be assumed.
Content	Optional. The attachment's content, in case it is directly embedded inside the message. For large attachments, it is RECOMMENDED that implementations make use of the "ExternalURI" attribute to refererence a copy of the content saved in an external storage mechanism.

5.8. The JavaScript Object Notation Serialization Method

This serialization method aims to convert IDMEFv2 messages to a format that is easy to parse and process, both by software/hardware processors, as well as humans. It relies on the the JavaScript Object Notation (JSON) Data Interchange Format defined in [RFC8259].

Conforming implementations MUST implement all the requirements specified in [RFC8259].

In addition, the following rules MUST be observed when serializing an IDMEFv2 message:

- The top-level Alert class (Section 5.2) is represented as a JSON object ([RFC8259]). This JSON object is returned to the calling process at the end of the serialization process.
- Aggregate classes are represented as JSON objects and stored as members of the top-level JSON object, using the same name as in the IDMEF data model. E.g. the appears under the name "Analyzer" inside the top-level JSON object.
- Attributes are stored as members of the JSON object representing the class they belong to, using the same name as in the IDMEF data model. E.g. the "Version" attribute from the is stored under the name "Version" inside the top-level JSON object.
- Lists from the IDMEF data model are represented as JSON arrays ([RFC8259]). This also applies to aggregate classes where a list is expected. E.g. the "Sensor" member inside the top-level JSON object contains a list of objects, where each object represents an instance of the .
- The various string-based data types listed in Section 3 are represented as JSON strings ([RFC8259]). Please note that the issues outlined in [RFC8259] regarding strings processing also apply here.
- IDMEF attributes with the "NUMBER" data type are represented as JSON numbers ([RFC8259]).

6. Security Considerations

This document describes a data representation for exchanging security-related information between incident detection system implementations. Although there are no security concerns directly applicable to the format of this data, the data itself may contain security-sensitive information whose confidentiality, integrity, and/or availability may need to be protected.

This suggests that the systems used to collect, transmit, process, and store this data should be protected against unauthorized use and that the data itself should be protected against unauthorized access.

The underlying messaging format and protocol used to exchange instances of the IDMEF MUST provide appropriate guarantees of confidentiality, integrity, and authenticity. The use of a standardized security protocol is encouraged.

The draft-lehmann-idmefv2-https-transport-01.txt document defines the transportation of IDMEF over HTTPs that provides such security.

7. IANA Considerations

This document creates 10 identically structured registries to be managed by IANA:

- Name of the registry group: "Incident Detection Message Exchange Format v2 (IDMEF)"
- URL of the registry: http://www.iana.org/assignments/idmefv2
- Namespace format: A registry entry consists of:
  - Rank. A uniq integer for this namespace. Range starts at 0 and ends at the length of this list. The maximum length of this list is 255.
  - Keyword. A keyword for a given IDMEF attribute. It MUST conform to the formatting specified by the IDMEF "ENUM" data type (Section 3.3.1).
  - Description. A short description of the enumerated keyword.
  - Reference. An optional list of URIs to further describe the value.
- Allocation policy: Expert Review per [RFC8126]. This reviewer will ensure that the requested registry entry conforms to the prescribed formatting. The reviewer will also ensure that the entry is an appropriate value for the attribute per the information model (Section 5).

The registries to be created are named in the "Registry Name" column of Table 5. Each registry is initially populated with ranks, keywords and descriptions that come from an attribute specified in the IDMEF model (Section 5). The initial Ranks, Keywords and Description fields of a given registry are listed in "Initial Values". The "Initial Values" column points to a table in this document that lists and describes each enumerated keyword. Each enumerated keyword in the table gets a corresponding entry in a given registry. The initial value of the Reference field of every registry entry described below should be this document.

Registry Name	**Initial Values**
Alert-Type	Table 1 (Alert class (Section 5.2))
Alert-Category	(Alert class (Section 5.2))
Alert-Cause	Table 2 (Alert class (Section 5.2))
Alert-Priority	Table 4 (Alert class (Section 5.2))
Alert-Status	Table 3 (Alert class (Section 5.2))
Analyzer-Category	(Alert class (Section 5.2))
Analyzer-Data	(Analyzer class (Section 5.3))
Analyzer-Method	(Analyzer class (Section 5.3))

Table 5: IANA Enumerated Value Registries

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- The H2020 7SHIELD project (Safety and Security Standards of Space Systems, ground Segments and Satellite data assets , via prevention, detection, response and mitigation of physical and cyber threats) who implemented in real scale first versions of IDMEFv2 on five pilots around Europe helping greatly to improve it.
- The CESNET team for their work on the [IDEA0] format (based on IDMEFv1) which inspired multiples concepts to IDMEFv2.

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Appendix A. Using IDMEFv2 in Ephemeral Environments (Informative)

Appendix B. Using IDMEFv2 in Ephemeral Environments (Informative)

The guidance in this appendix is informative. It describes implementation practices for representing resources whose lifetime may be shorter than the retention period of the alerts and telemetry associated with them. It does not change the IDMEFv2 data model or the definition of any IDMEFv2 attribute.

The examples are illustrative and are not complete IDMEFv2 messages unless stated otherwise. Platform-specific reference formats shown below are local implementation conventions; they are not URI schemes standardized by IDMEFv2.

The key words , , , , and in this appendix are to be interpreted as described in . Because this appendix is informative, these words describe recommended implementation behavior in the stated context and do not introduce additional requirements for conformance to the IDMEFv2 data model.

B.1. The Ephemeral Asset Problem

Container orchestration platforms and cloud runtime environments create and terminate resources dynamically. IP addresses may change or be reassigned, resource names may be reused, and a resource involved in a security incident may no longer exist when an investigation takes place.

Consequently, an IP address or hostname alone may be insufficient for reliable incident correlation and post-incident investigation. Implementations can combine the IDMEFv2 target identifier with available network, operational, and infrastructure context to make alerts easier to correlate over time.

The attributes discussed in this appendix serve different purposes:

- identifies the target in the IDMEFv2 data model.
- records an IP address associated with the target in the context of the event.
- records an operational hostname or resource name.
- describes the target's location according to the IDMEFv2 attribute definition. Some implementations may use it, by documented convention, to carry a platform-specific resource reference.

These attributes are complementary; none of them, by itself, guarantees that historical metadata or the resource itself will remain available after the resource terminates.

B.2. Identification and Correlation

B.2.1. Target.ID

is the identifier of the target within the IDMEFv2 data model and MUST be populated as specified by the applicable IDMEFv2 version. It should not automatically be treated as the native identifier assigned by a cloud provider or orchestrator: the two identifiers can have different scopes and lifetimes.

Implementations that correlate multiple alerts about the same resource SHOULD document how values of are assigned and how they map to platform-native resource identifiers. When a platform reuses an operational name, implementations SHOULD distinguish successive resource instances when that distinction is needed for the intended correlation and investigation use case.

A value does not, by itself, guarantee access to the resource or to historical metadata after termination. Implementations should preserve the mapping between IDMEFv2 identifiers and native identifiers for as long as the relevant investigation requirements apply.

B.2.2. Target.IP

records the network address associated with the target in the context of the event. It can support correlation with network telemetry, firewall logs, flow records, and other monitoring data. An IP address is a time-dependent network attribute and should not be treated as a persistent identifier of an ephemeral resource.

When available, the event timestamp and relevant network context should be retained to support interpretation of the address at the time of the incident. Implementations should distinguish addresses belonging to different infrastructure layers, such as a pod, container, node, NAT gateway, or proxy, when that distinction matters to the investigation.

B.2.3. Target.Hostname

records the operational hostname or resource name observed for the target. Examples include a Kubernetes pod name, a container name, a Nomad allocation-related name, or a runtime-specific hostname.

A hostname can help identify and correlate a resource during its lifetime. However, names may be reused, and their uniqueness may be limited to a namespace, cluster, runtime, or time interval. A hostname should therefore not be considered globally unique or permanently stable without additional context.

B.2.4. Target.Location and Platform-Specific References

The meaning of remains the meaning defined by the applicable IDMEFv2 specification. Using this attribute to carry a platform-specific resource reference is an implementation convention; it does not redefine the attribute or establish a common reference syntax for all IDMEFv2 implementations.

Where an implementation adopts this convention, the value may contain a platform resource identifier, an orchestration resource path, a container identifier qualified by its runtime, or a documented combination of identifiers. Implementations SHOULD prefer a reference to the individual resource involved in the event rather than a reference only to its parent cluster, service, or workload definition, when individual-instance identification is needed.

Implementations using platform-specific references SHOULD document at least:

- the syntax and interpretation of the reference;
- the scope in which the reference is unique;
- whether the reference identifies a resource instance, a configuration, or a parent resource;
- the expected stability and lifetime of the reference; and
- how the reference is associated with event time and retained historical records.

A value does not, by itself, guarantee that a terminated resource can be retrieved or that its historical metadata remains available. Systems that require post-incident traceability need to retain the relevant metadata and security telemetry in their surrounding monitoring and logging infrastructure.

B.3. Platform-Specific Examples

The following examples illustrate possible implementation conventions. The schemes , , , , , and are illustrative labels only. They are not defined or standardized by IDMEFv2, and the examples do not imply that these schemes are registered URI schemes or official resource URI formats for the named platforms.

The values are fictitious examples. In a deployment, implementations should use identifiers obtained from the relevant platform and follow the platform's rules for identifying the resource concerned.

B.3.1. Kubernetes

A Kubernetes pod name may not be sufficient to distinguish successive pod instances. When available and relevant, the pod UID can be retained along with the cluster and namespace context. Container-specific information may also be included when the incident concerns an individual container.

Example:

```
{
  "Target": [
    {
      "ID": "7d6a3c21-1b4e-4f80-9a12-6e5b2c8d901f",
      "IP": "10.244.2.15",
      "Hostname": "payment-worker-5c8d9f-abc123",
      "Location": "k8s://cluster-prod-eu/namespace-paiement/
pod/payment-worker-5c8d9f-abc123?uid=6a1f8b42-3c9d-4e70-
a521-8b7d9c2e104f&container=worker&node=node-worker-03",
      "Service": "finance"
    }
  ]
}
```

The value is an illustrative convention. The pod UID identifies a Kubernetes object instance; cluster and namespace context help interpret the reference. Node-level information may be included when useful for correlation with network sensors or host-level telemetry.

B.3.2. Google Kubernetes Engine (GKE)

For GKE, an implementation may combine project and cluster context with the Kubernetes namespace, pod name, and pod UID. A cluster resource identifier alone identifies the cluster, not the individual pod involved in the incident.

Example:

```
{
  "Target": [
    {
      "ID": "8e7b4d32-2c5f-4a91-8b23-7f6c3d9e012a",
      "IP": "10.244.2.15",
      "Hostname": "payment-worker-5c8d9f-abc123",
      "Location": "gke://projects/my-project-123/locations/us-
west1/clusters/cluster-prod-eu/namespaces/namespace-paiement/
pods/payment-worker-5c8d9f-abc123?uid=6a1f8b42-3c9d-4e70-
a521-8b7d9c2e104f&container=worker",
      "Service": "finance"
    }
  ]
}
```

The value is an illustrative convention, not an official Google Cloud resource URI format. An implementation may instead record a documented Google Cloud resource identifier together with the Kubernetes identifiers needed to identify the individual pod.

B.3.3. Amazon Elastic Container Service (ECS)

In Amazon ECS, a task ARN identifies a task instance, whereas a task definition ARN identifies a task definition revision. When investigating a particular task execution, the reference should identify the task instance rather than only its task definition. The task definition ARN may be retained separately as configuration context.

Example:

```
{
  "Target": [
    {
      "ID": "9f8c5e43-3d60-4ba2-9c34-8a7d4e0f123b",
      "IP": "10.0.1.42",
      "Hostname": "payment-worker-5c8d9f-abc123",
      "Location": "arn:aws:ecs:us-west-1:123456789012:task/cluster-prod-
eu/0123456789abcdef0123456789abcdef",
      "Service": "finance"
    }
  ]
}
```

The ARN is illustrative. A deployment should use the actual task ARN returned by ECS and retain sufficient cluster context to interpret the reference when needed. If the incident concerns an individual container within a task, the relevant container identifier may also be retained. A task definition ARN should not be substituted for a task-instance identifier when distinguishing individual executions is required.

B.3.4. Azure Kubernetes Service (AKS)

In AKS, the Azure Resource Manager (ARM) resource ID of a managed cluster identifies the cluster, not an individual pod. A pod-level reference may combine subscription, resource group, and cluster context with the Kubernetes namespace, pod name, and pod UID.

Example:

```
{
  "Target": [
    {
      "ID": "a09d6f54-4e71-4cb3-8d45-9b8e5f10234c",
      "IP": "10.240.0.15",
      "Hostname": "payment-worker-5c8d9f-abc123",
      "Location": "aks://
subscriptions/12345678-1234-1234-1234-123456789012/resourceGroups/my-
rg/managedClusters/cluster-prod-eu/namespaces/namespace-paiement/pods/
payment-worker-5c8d9f-abc123?uid=6a1f8b42-3c9d-4e70-a521-8b7d9c2e104f",
      "Service": "finance"
    }
  ]
}
```

The value is an illustrative convention, not an official ARM resource identifier for a pod. An implementation may retain the cluster ARM resource ID separately or include it in a documented platform-specific representation. The reference should distinguish the individual Kubernetes resource from its parent cluster when that distinction is required.

B.3.5. HashiCorp Nomad

In Nomad, an allocation identifier can refer to a particular workload allocation. The reference should include sufficient context to distinguish allocations across clusters, regions, namespaces, or jobs, as applicable.

Example:

```
{
  "Target": [
    {
      "ID": "b10e7a65-5f82-4dc4-9e56-0c9f6a21345d",
      "IP": "172.17.0.5",
      "Hostname": "payment-worker-5c8d9f-abc123",
      "Location": "nomad://region-eu/cluster-prod/job-payment/
alloc-01234567-89ab-cdef-0123-456789abcdef?task=worker",
      "Service": "finance"
    }
  ]
}
```

The value is an illustrative convention. Implementations should use the allocation identifier supplied by Nomad and document any context needed to interpret it. Task-level or container-level information may be included when necessary to distinguish the runtime entity involved.

B.3.6. Docker and Podman

For standalone container runtimes such as Docker and Podman, a container name may be insufficient because names can be reused after a container is removed. A reference should identify the container instance and, where necessary, the runtime host or engine. The full container identifier is generally preferable to a human-readable name.

Example:

```
{
  "Target": [
    {
      "ID": "c21f8b76-6083-4ed5-af67-1d0a7b32456e",
      "IP": "172.17.0.3",
      "Hostname": "web-app-1",
      "Location": "docker://server-web-01/containers/
a1b2c3d4e5f67890123456789012345678901234567890123456789012345678",
      "Service": "e-commerce"
    }
  ]
}
```

The value is an illustrative convention, not an official Docker URI scheme. A corresponding convention may be defined for Podman, for example using . Implementations should document whether the reference identifies the container, runtime instance, host, or a combination of these entities. Where multiple runtime hosts or engines are involved, the reference should include enough context to avoid ambiguity.

B.4. Service Attribute

The attribute should be used consistently with the attribute definition in the applicable IDMEFv2 specification. Values such as , , and may identify high-level business services associated with a target.

Implementations should avoid using as an undocumented substitute for platform-specific resource identifiers, such as pod names, container IDs, allocation IDs, task ARNs, or cluster resource IDs. Where business services and technical runtime services must be distinguished, implementations should document the mapping between those concepts and the relevant IDMEFv2 attributes.

B.5. Temporal Context and Historical Retention

Reliable post-incident investigation requires more than a resource reference. Implementations should preserve the event timestamp and sufficient context to determine which resource instance, network address, and runtime environment were involved at the time of the event.

Depending on the environment and investigation requirements, relevant records may include orchestration events, container lifecycle records, cloud audit logs, network flow records, and mappings between resource identifiers and network addresses. Systems that require long-term traceability need to retain the historical metadata and security telemetry needed for the intended investigation period.

A value may be used as a lookup key into retained historical records when the implementation has documented that convention. The availability and retention of those records are properties of the surrounding monitoring and logging infrastructure, not guarantees provided by the attribute itself.

B.6. Summary

Ephemeral environments require a distinction between target identification, network correlation, and historical resource context. remains the target identifier defined by IDMEFv2; records a network address associated with the event; provides an operational name; and retains the meaning assigned to it by the applicable IDMEFv2 specification, although implementations may document a convention for carrying a platform-specific reference in that attribute.

For reliable correlation, implementations should distinguish successive resource instances when required, retain sufficient temporal and infrastructure context, and document the syntax and scope of platform-specific references. Long-term traceability also depends on the retention and availability of historical metadata and security records. These practices allow IDMEFv2 to be used with ephemeral resources without requiring a separate IDMEFv2 extension for every orchestration platform or container runtime.

Appendix C. Enumerations

C.1. Entity Sector

Rank	Keyword	Definition
0	Undefined	Sector undefined
1	Administration.CentralGovernment	Central government ministries and agencies
2	Administration.Diplomatic	Embassies, consulates, and diplomatic missions
3	Administration.EmergencyServices	Police, fire, rescue, and emergency medical services
4	Administration.Judiciary	Courts, judicial systems, and correctional facilities
5	Administration.Local	Regional and municipal public services
6	Administration.Other	Other public administration entities critical for the continuity of public services
7	Banking.Banking	Banking institutions and credit activities
8	Banking.Other	Other banking and financial entities critical for the stability of the financial system
9	Cemeteries.Crematoria	Crematoria operations
10	Cemeteries.Public	Public cemetery and crematorium facilities
11	Cemeteries.Other	Other cemetery and funeral service entities critical for public health and social continuity
12	Chemical.Production	Production of hazardous and high-consequence chemicals
13	Chemical.Storage	Chemical storage and distribution facilities
14	Chemical.Other	Other chemical sector entities critical for safety and security
15	Civil.ElectionMonitoring	Election observation and monitoring entities
16	Civil.HumanitarianAid	Humanitarian aid organizations
17	Civil.NGOs	Non-governmental organizations with critical functions
18	Civil.Other	Other civil society entities critical for democratic processes and social stability
19	Cultural.Archives	National and regional archives
20	Cultural.HistoricalSites	Historical monuments and archaeological sites
21	Cultural.Libraries	National and public libraries
22	Cultural.Museums	Museums and exhibition spaces

Rank	Keyword	Definition
23	Cultural.Other	Other cultural heritage entities critical for national identity and continuity
24	Defense.CommandControl	Military C4ISR systems (command, control, communications, computers, intelligence, surveillance, reconnaissance)
25	Defense.Cyber	Military cyber defense and security operations centers
26	Defense.IndustrialBase	Defense manufacturing, weapons systems, munitions production
27	Defense.Logistics	Military supply chains, fuel depots, ammunition storage
28	Defense.Installations	Military bases, headquarters, and operational facilities
29	Defense.Research	Defense laboratories and R&D facilities
30	Defense.Other	Other defense and military entities critical for national security
31	Digital.CloudServices	Cloud computing and critical digital services
32	Digital.DataCenters	Data center hosting and infrastructure services
33	Digital.DigitalProviders	Online marketplaces, search engines and social media platforms
34	Digital.DomainNameSystems	DNS service providers and TLD registries
35	Digital.ICTServiceManagement	ICT service management B2B services
36	Digital.SatelliteCommunications	Satellite communication networks and ground stations
37	Digital.TelecomNetworks	Fixed and mobile telecommunications networks
38	Digital.UnderseaCables	Submarine communications cable infrastructure
39	Digital.Other	Other digital and telecommunications entities critical for the continuity of digital services
40	Education.PrimarySecondary	Primary and secondary schools
41	Education.ResearchSchools	Research-focused educational institutions
42	Education.Universities	Universities and higher education institutions
43	Education.Other	Other educational entities critical for societal continuity
44	Emergency.CivilProtection	Civil protection and disaster management agencies

Rank	Keyword	Definition
45	Emergency.Medical	Ambulance services and emergency medical response
46	Emergency.FireAndRescue	Firefighting and rescue services
47	Emergency.Police	Law enforcement agencies and operations
48	Emergency.Other	Other emergency response entities critical for public safety
49	Energy.DistrictHeating	Operation of district heating networks
50	Energy.Electricity	Generation, transmission and distribution of electrical power
51	Energy.Gas	Transport, storage and distribution of natural gas
52	Energy.Hydrogen	Production, transport and storage of hydrogen
53	Energy.Nuclear	Nuclear power generation and fuel cycle facilities
54	Energy.Oil	Refining, transport and storage of petroleum products
55	Energy.Other	Other energy entities critical for the continuity of energy supply
56	Finance.MarketInfrastructures	Stock exchanges and clearing houses
57	Finance.Insurance	Systemically important insurance entities
58	Finance.PaymentSystems	Card schemes, transfers and instant payment systems
59	Finance.Other	Other financial entities critical for the stability of the financial system
60	Food.AgriculturalProduction	Strategic agricultural and livestock production
61	Food.FoodDistribution	Retail and logistics for food supply chains
62	Food.FoodProcessing	Industrial food processing and manufacturing
63	Food.FoodSafety	Food safety inspection and control authorities
64	Food.Other	Other food sector entities critical for the continuity of the food supply chain
65	Health.BloodAndTissue	Blood banks, tissue banks, and transplantation services
66	Health.HealthcareProviders	Hospitals and clinics providing essential care
67	Health.Laboratories	Medical analysis and diagnostic laboratories
68	Health.MedicalDevices	Manufacture and maintenance of critical medical devices
69	Health.PharmaceuticalSupplyChain	Manufacturing, wholesale and distribution of pharmaceuticals
70	Health.PublicHealth	Public health agencies and epidemiological surveillance

Rank	Keyword	Definition
71	Health.Other	Other health entities critical for the continuity of healthcare services
72	Logistics.FreightForwarding	Freight forwarding and cargo management
73	Logistics.LastMileDelivery	Last-mile delivery services
74	Logistics.ThirdPartyLogistics	Third-party logistics providers
75	Logistics.Warehousing	Strategic warehousing and storage facilities
76	Logistics.Other	Other logistics entities critical for supply chain continuity
77	Manufacturing.Aerospace	Aerospace and aviation manufacturing
78	Manufacturing.Batteries	Battery manufacturing and energy storage production
79	Manufacturing.Chemical	Chemical manufacturing and industrial processes
80	Manufacturing.Electronic	Manufacture of computer, electronic and optical products
81	Manufacturing.Defense	Defense manufacturing and strategic military supply chains
82	Manufacturing.MedicalDevices	Manufacture of medical devices and equipment
83	Manufacturing.MotorVehicles	Manufacture of motor vehicles and transport equipment
84	Manufacturing.Pharmaceutical	Pharmaceutical manufacturing and active pharmaceutical ingredients
85	Manufacturing.Semiconductors	Semiconductor fabrication and microelectronics manufacturing
86	Manufacturing.Strategic	Critical manufacturing such as metallurgy and components
87	Manufacturing.Other	Other manufacturing entities critical for industrial resilience and strategic supply chains
88	Media.Infrastructure	Transmitters, towers, and broadcast distribution infrastructure
89	Media.OnlineMedia	Digital news platforms and content providers
90	Media.Press	Newspapers, publishing houses, and press agencies
91	Media.Radio	Radio broadcasting networks and studios
92	Media.Television	Television broadcasting networks and studios
93	Media.Other	Other media entities critical for information integrity and public communication
94	Mining.CriticalRawMaterials	Extraction and processing of critical raw materials
95	Mining.EnergyMinerals	Coal, uranium, and other energy mineral mining

Rank	Keyword	Definition
96	Mining.Other	Other mining entities critical for resource security
97	Nuclear.FuelCycle	Nuclear fuel production, enrichment, reprocessing, and waste management
98	Nuclear.Medical	Nuclear medical facilities and radioisotope production
99	Nuclear.PowerGeneration	Civil nuclear power plants and associated facilities
100	Nuclear.Research	Nuclear research reactors and laboratories
101	Nuclear.Other	Other nuclear entities critical for safety and security
102	Postal.CourierServices	Postal and courier services
103	Postal.Other	Other postal and courier entities critical for the continuity of mail and parcel services
104	Religious.PilgrimageSites	Major pilgrimage destinations
105	Religious.PlacesOfWorship	Churches, mosques, synagogues, temples, and other religious buildings
106	Religious.Other	Other religious sites critical for community continuity
107	Research.BiologicalSafety	BSL-3 and BSL-4 high-containment laboratories
108	Research.ChemicalSafety	High-containment chemical research facilities
109	Research.Research	Key research laboratories with strategic importance
110	Research.Other	Other research entities critical for strategic research continuity and innovation
111	Space.GroundStations	Satellite ground control and telemetry stations
112	Space.LaunchFacilities	Space launch sites and associated infrastructure
113	Space.SpaceActivities	Space industry and satellite operations
114	Space.Other	Other space entities critical for the continuity of space-based services and infrastructure
115	Transport.Aviation	Airports, air traffic control and airline operations
116	Transport.Maritime	Ports, terminals and maritime traffic management
117	Transport.Pipeline	Oil, gas, and hydrogen pipeline infrastructure
118	Transport.PublicTransport	Urban and regional public transportation systems
119	Transport.Rail	Railway infrastructure and train operations
120	Transport.Road	Traffic management and strategic road logistics

Rank	Keyword	Definition
121	Transport.Other	Other transport entities critical for the continuity of passenger and freight mobility
122	Waste.Hazardous	Collection, treatment and disposal of hazardous waste
123	Waste.NonHazardous	Management of non-hazardous solid waste
124	Waste.Nuclear	Nuclear waste management and storage facilities
125	Waste.Recycling	Waste processing and recycling operations
126	Waste.Other	Other waste management entities critical for the continuity of waste services
127	Water.DamsAndReservoirs	Dam and reservoir infrastructure for water management
128	Water.DrinkingWater	Capture, treatment and distribution of potable water
129	Water.Irrigation	Large-scale agricultural irrigation systems
130	Water.Wastewater	Collection and treatment of sewage and wastewater
131	Water.Other	Other water entities critical for the continuity of water supply and sanitation

Table 6: enum_alert_entitysector

C.2. Alert Category

uank	Keyword	Description
0	Abuse.Coercion	The practice of compelling an individual to act against their will by using force, threats, intimidation, or extreme dependency, often for personal or financial gain.
1	Abuse.Extermism	The process of socializing an individual, often through psychological manipulation, to adopt the beliefs and goals of a violent extremist group and become willing to engage in activities that support the group.
2	Abuse.Grooming	The process of deliberately building an emotional connection with a person to lower their inhibitions for the purpose of sexual abuse, exploitation, or trafficking.
3	Abuse.Harassment	A pattern of unwanted, intrusive behavior (physical,

uank	Keyword	Description
		verbal, or online) directed at a specific person that causes fear, distress, or emotional harm.
4	Abuse.Trafficking	The act of recruiting, transporting, transferring, harboring, or receiving a person through force, fraud, or coercion for the purpose of exploitation, such as forced labor or sexual servitude.
5	Abuse.Other	Any other incident involving manipulation or coercion of people for harmful purposes that does not fit into the specific abuse subcategories.
6	Access.Authorized	An incident involving legitimate, approved access to resources that may be relevant for auditing, monitoring, or establishing a baseline of normal activity.
7	Access.Backdoor	The installation or discovery of a concealed method of bypassing normal authentication or encryption in a computer system, product, or embedded device.
8	Access.Clonned	The act of creating an unauthorized copy of a physical or digital security credential, such as an access card, to gain illicit entry to a facility or system.
9	Access.Compromise	An incident where a legitimate user's credentials (e.g., username/password) are stolen or guessed and used by an unauthorized individual to gain access to systems or data.
10	Access.Escalation	An incident where a user or process gains access rights, permissions, or capabilities that exceed those normally assigned, often to bypass security restrictions.
11	Access.Forced	An incident involving the use of physical force to breach a barrier (e.g., lock, door, window) or the repeated systematic guessing of passwords to gain unauthorized access (bruteforce attack).

uank	Keyword	Description
12	Access.Lost	An incident where physical or digital access credentials (e.g., keys, ID badges, login details) are misplaced, stolen, or used by an unauthorized person.
13	Access.Tailgating	A physical security breach where an unauthorized person follows an authorized individual through a secured entry point, circumventing access control mechanisms.
14	Access.Unauthorized	Incident involving unauthorized access — physical or digital — to a restricted building, room, area, system, network, application, account, or data resource, whether actual or attempted.
15	Access.Other	Any other incident related to physical or digital access that does not fit into the specific access subcategories.
16	Availability.DDoS	An incident where multiple compromised systems (a botnet) are used to target a single system with a flood of traffic, causing a denial of service.
17	Availability.DoS	An incident where a single machine or network attempts to make a system, service, or network resource unavailable by overwhelming it with malicious requests or traffic.
18	Availability.Failure	An incident caused by the unintentional malfunction of hardware or software due to errors, bugs, wear and tear, or other faults, leading to service degradation or unavailability.
19	Availability.HeartBeat	A periodic signal generated by hardware or software to indicate normal operation, often used for monitoring system health, connectivity, or location tracking.
20	Availability.Misconfiguration	An incident where incorrect configuration of systems, software, or networks leads to service disruptions, outages, or security vulnerabilities.
21	Availability.Outage	An incident where essential utilities or services (such

uank	Keyword	Description
		as electricity, water, or network connectivity) become unavailable, disrupting normal operations.
22	Availability.Overload	An incident where a system or component is subjected to a load beyond its designed capacity, leading to performance degradation or failure, even if the load is not malicious.
23	Availability.Other	Any other incident that impacts the availability of resources or services, not covered by the specific subcategories.
24	Biological.Animal	An incident where animals pose a direct threat to human safety, public health (e.g., zoonotic diseases), or economic stability (e.g., livestock diseases).
25	Biological.Epidemic	The rapid and widespread occurrence of an infectious disease in a specific population or region, exceeding what is normally expected.
26	Biological.Insect	An incident involving a harmful outbreak or infestation of insects that threatens public health, agriculture, livestock, or property.
27	Biological.Zombies	A fictional or hypothetical scenario involving a pathogen that causes a pandemic of aggressive, infectious behavior, often used as a metaphor for worst-case outbreak scenarios in planning.
28	Biological.Other	Any other incident caused bybiological not covered by specific subcategories.
29	Climat.Drought	A prolonged period of below-average precipitation leading to a water shortage, which can impact agriculture, ecosystems, and water supplies.
30	Climat.LakeOutburst	An incident where a glacial lake dam (often moraine or ice) fails, rapidly releasing a large volume of water and causing devastating floods downstream.
31	Climat.Wildfire	An unplanned and uncontrolled fire burning in natural or rural areas, often exacerbated

uank	Keyword	Description
		by climatic conditions like drought, wind, and heat.
32	Climat.Other	Any other incident caused by long-lived atmospheric processes (climatological) not covered by specific subcategories.
33	Extraterrestrial.Aliens	A hypothetical incident involving the discovery of or interaction with extraterrestrial intelligent life, a theoretical scenario in scientific and security planning.
34	Extraterrestrial.Impact	An incident involving a celestial object (asteroid, meteoroid, comet) colliding with Earth, potentially causing localized or global damage.
35	Extraterrestrial.SpaceWeather	An event caused by solar or cosmic activity, such as solar flares or geomagnetic storms, that can disrupt Earth's technological infrastructure.
36	Extraterrestrial.Other	Any other incident caused extraterrestrial process not covered by specific subcategories.
37	Fraud.Copyright	The act of reproducing, distributing, or installing software, media, or other materials in violation of their copyright, often for personal gain or distribution (piracy).
38	Fraud.Corruption	A fraudulent scheme that is made possible by the abuse of power or position by a trusted individual (e.g., employee, official) who acts for personal gain.
39	Fraud.Espionnage	The use of illegal or unethical means, such as hacking, bribery, or theft, to acquire a competitor's trade secrets, intellectual property, or other confidential business information.
40	Fraud.Masquerade	A type of attack where an attacker illegitimately assumes the identity of another user, process, or system to gain unauthorized access, privileges, or benefits.

uank	Keyword	Description
41	Fraud.Phishing	A cyber attack where an attacker disguises themselves as a trustworthy entity (e.g., via email or fake website) to trick a victim into revealing sensitive information like usernames, passwords, or credit card details.
42	Fraud.Usage	The use of an organization's assets (e.g., computing power, network, email) for non-work-related, often illegal, activities without authorization.
43	Fraud.Other	Any other incident involving deception for financial or reputational gain that does not fit into the specific fraud subcategories.
44	Geophysical.Earthquake	An incident caused by a sudden, rapid shaking of the earth resulting from the movement of tectonic plates, which can cause ground shaking, surface rupture, and tsunamis.
45	Geophysical.MassMovement	An incident involving the downslope movement of rock, soil, or snow under the force of gravity, such as landslides, avalanches, or rockfalls.
46	Geophysical.Other	Any other incident caused by solid-earth processes not covered by specific geophysical subcategories.
47	Geophysical.Volcanic	An incident caused by the eruption of magma and volcanic gases from a volcano, which can produce lava flows, pyroclastic flows, ashfall, and lahars.
48	Hydro.Flood	An incident where water submerges land that is normally dry, often caused by heavy rain, storm surge, or dam failure, leading to property damage and risk to life.
49	Hydro.Landslide	An incident involving the downward movement of slope materials (soil, rock) triggered by water saturation from heavy rain or snowmelt.
50	Hydro.Wave	An incident involving destructive waves, such as

uank	Keyword	Description
		tsunamis or storm surges, that can cause coastal flooding, erosion, and damage.
51	Hydro.Other	Any other incident caused by the movement, distribution, and quality of water, not covered by specific hydrological subcategories.
52	Insider.Malicious	A security incident caused by a current or former employee, contractor, or other trusted insider who intentionally acts to harm the organization, its data, or its people.
53	Insider.Negligent	A security incident caused unintentionally by an insider, such as through carelessness, lack of awareness, or simple human error, leading to data exposure or system compromise.
54	Insider.Other	Any other security incident involving an insider (trusted individual) that does not fit into the specific insider threat subcategories.
55	Malware.Adware	A type of software that automatically displays or downloads unwanted advertisements, often in a disruptive manner, and may track user behavior.
56	Malware.Backdoor	A type of malware that bypasses normal authentication to give an attacker persistent remote access to a compromised system, often enabling data theft or installation of additional malware.
57	Malware.Cryptominer	A type of malware that secretly uses a victim's computing resources to mine cryptocurrency, causing degraded performance and increased power consumption without consent.
58	Malware.Downloader	A type of malware designed to download and install other malicious software from a remote server, typically acting as the first stage in a multi-step attack.

uank	Keyword	Description
59	Malware.Ransomware	A type of malware that encrypts a victim's files or systems, rendering them inaccessible, and demands a ransom payment, often in cryptocurrency, for the decryption key.
60	Malware.Rootkit	A type of malware designed to hide its presence and grant an attacker persistent, privileged access to a compromised system while evading detection.
61	Malware.Spyware	A type of software that secretly monitors and collects information about a user's activities, such as keystrokes, browsing habits, and personal data, without their consent.
62	Malware.Trojan	A type of malware that disguises itself as legitimate or desirable software to trick users into installing it, after which it can perform malicious actions.
63	Malware.Virus	A type of malicious software that attaches itself to a legitimate program or file and replicates itself to spread to other programs, often requiring human action to propagate.
64	Malware.Worm	A type of standalone malicious software that replicates itself to spread across networks, often exploiting vulnerabilities without requiring human interaction.
65	Malware.Other	Any other incident involving malicious software that does not fit into the specific malware subcategories.
66	Meteo.Cold	A meteorological incident involving a rapid fall in temperature over a short period, or a prolonged period of extreme cold, posing risks to health and infrastructure.
67	Meteo.Fog	A meteorological incident where dense fog reduces visibility, potentially disrupting transportation and causing accidents.
68	Meteo.Heat	A meteorological incident involving a prolonged period of excessively hot weather,

uank	Keyword	Description
		which can cause health impacts, infrastructure stress, and environmental damage.
69	Meteo.Rain	A meteorological incident involving excessive or prolonged rainfall that can lead to flooding, landslides, and transportation disruptions.
70	Meteo.Snow	A meteorological incident involving heavy snowfall and blizzard conditions, which can disrupt transportation, damage infrastructure, and pose risks to safety.
71	Meteo.Wind	A meteorological incident involving damaging or dangerous winds, such as from storms, tornadoes, or hurricanes, that can cause structural damage and power outages.
72	Meteo.Other	Any other incident caused by atmospheric processes not covered by specific meteorological subcategories.
73	National.Conflict	A state of armed conflict between two or more nations, involving organized military forces and impacting national security.
74	National.Crime	Large-scale illegal activities, such as trafficking in drugs, weapons, or people, conducted by sophisticated criminal networks that pose a threat to national and international security.
75	National.Cyber	Large-scale cyber operations, including attacks, espionage, and disinformation campaigns, conducted by state-sponsored or state-affiliated groups against national interests.
76	National.Economical	An incident involving significant disruption to a nation's economy, such as market crashes, hyperinflation, or trade wars, affecting national stability.
77	National.Environemental	Incidents related to large-scale environmental shifts, such as climate change, resource scarcity, or global pandemics,

uank	Keyword	Description
		that have significant national security implications.
78	National.Societal	An incident involving large-scale public disorder, such as strikes, riots, or civil unrest, that challenges social order and may require national-level response.
79	National.Terrorism	An incident involving the use of violence, intimidation, or threats by non-state actors against civilians or property to achieve political, religious, or ideological objectives.
80	National.Other	Any other incident that has a significant impact at the national scale, not covered by specific national security subcategories.
81	Operational.Misuse	The use of an organization's assets (e.g., computers, network, time) for purposes that are not officially authorized, which may violate policy but not be malicious.
82	Operational.Policy Violation	An incident where an individual fails to comply with established organizational policies, procedures, or security rules, whether intentionally or accidentally.
83	Operational.Process Failure	An incident where a designed process, procedure, or workflow fails to achieve its intended outcome, leading to operational or security gaps.
84	Operational.Other	Any other operational issue that does not fit into the specific operational incident subcategories.
85	Recon.Aerial	The act of using aerial platforms, such as drones or satellites, to conduct surveillance or gather intelligence about a target area.
86	Recon.Landscape	The act of physically observing a location to gather information about security measures, layouts, or potential vulnerabilities.
87	Recon.Network	The practice of probing a network to discover active hosts, open ports, and services,

uank	Keyword	Description
		often as a precursor to an attack or unauthorized monitoring.
88	Recon.OSINT	The practice of collecting and analyzing information from publicly available sources (e.g., social media, public records) for intelligence purposes.
89	Recon.Other	Any other incident involving scanning, spying, or monitoring to identify resources that does not fit into specific recon subcategories.
90	Sabotage.Data	The intentional act of deleting, altering, or corrupting digital or physical data to cause harm, disrupt operations, or cover tracks.
91	Sabotage.Destruction	The intentional and malicious act of destroying or severely damaging physical assets, such as windows, equipment, or buildings.
92	Sabotage.Disruption	The intentional disruption of essential services or utilities, such as electricity, water, or network connectivity, to cause operational downtime.
93	Sabotage.Equipment	The intentional act of causing damage to operational equipment, machinery, or vehicles, often to disrupt production or operations.
94	Sabotage.Graffiti	The act of willfully defacing, damaging, or marking public or private property with inscriptions, drawings, or tags without permission.
95	Sabotage.Tampering	The act of deliberately meddling with or disabling security controls (e.g., locks, alarms, cameras) to compromise their effectiveness.
96	Sabotage.Vandalism	The deliberate and malicious act of damaging, destroying, or obstructing an organization's physical assets, operations, or systems.
97	Sabotage.Other	Any other incident involving the intentional damage to property or assets not covered by the specific sabotage subcategories.

uank	Keyword	Description
98	Safety.Accident	An unplanned, unforeseen event (e.g., vehicle crash, industrial mishap, chemical spill) that results in injury, loss of life, or damage to health.
99	Safety.Agression	An incident where an individual uses physical force against another person, causing bodily harm, pain, or the fear of immediate harm.
100	Safety.Explosion	A sudden, violent release of energy (e.g., from gas, chemicals, or explosives) that causes a blast, fire, and potential structural damage, injury, or loss of life.
101	Safety.Fire	An incident involving uncontrolled burning (e.g., structural, wildland, or chemical fire) that threatens human safety, property, or the environment.
102	Safety.Hostage	An incident where a person or group is held against their will by a captor, often to compel a third party to meet certain demands.
103	Safety.Sexual	An incident involving any unwanted sexual act, contact, or behavior directed against an individual without their consent.
104	Safety.Other	Any other incident that causes or has the potential to cause injury, loss of life, or endanger citizens, not covered by specific safety subcategories.
105	SocialEng.Baiting	A social engineering attack that lures victims by offering something desirable (e.g., free music, a prize) in exchange for information or by tricking them into downloading malware.
106	SocialEng.Phishing	A type of social engineering where attackers send fraudulent emails, appearing to be from a legitimate source, to trick recipients into revealing sensitive information or installing malware.
107	SocialEng.Pretexting	A social engineering tactic where the attacker creates a fabricated scenario or pretends

uank	Keyword	Description
		to be someone they are not to engage a victim and extract information or access.
108	SocialEng.QuidProQuo	A social engineering tactic where the attacker offers a service or benefit (e.g., technical support) in exchange for information or access, often expecting something in return later.
109	SocialEng.Smishing	A social engineering attack conducted via SMS (text message), where the attacker sends a fraudulent message to trick the recipient into clicking a malicious link or providing information.
110	SocialEng.Spear Phishing	A highly targeted phishing attack directed at a specific individual, organization, or role, often using personalized information to increase credibility.
111	SocialEng.Vishing	A social engineering attack conducted over the phone, where the attacker impersonates a legitimate entity to trick the victim into revealing sensitive information or performing actions.
112	SupplyChain.Compromise	A security incident where an attacker exploits a vulnerability in a third-party vendor's system to gain access to or compromise the primary target's network or data.
113	SupplyChain.Disruption	An event that disrupts the normal flow of products, services, or information within a supply chain, often impacting operations and delivery.
114	SupplyChain.Other	Any other incident affecting the supply chain that does not fit into the specific supply chain subcategories.
115	SocialEng.Other	Any other incident involving psychological manipulation of people to divulge information or perform actions, not covered by specific social engineering subcategories.
116	Theft.Breaches	A security incident involving the unauthorized access,

uank	Keyword	Description
		acquisition, or disclosure of sensitive, protected, or confidential data.
117	Theft.Data	The unauthorized taking or copying of sensitive or confidential documents, whether in physical or digital form.
118	Theft.Equiment	The unlawful taking of physical hardware, such as computers, mobile phones, or servers, resulting in loss of assets and potentially the data they contain.
119	Theft.FinInfo	The theft of financial information, such as credit card numbers or bank account details, often for the purpose of fraudulent transactions.
120	Theft.IP	The unlawful acquisition of a company's intellectual property, including trade secrets, patents, formulas, or proprietary processes.
121	Theft.Machinery	The unlawful taking of heavy equipment, vehicles, or specialized machinery, often resulting in significant operational and financial loss.
122	Theft.PII	The unauthorized acquisition of Personally Identifiable Information (PII) that can be used to identify, contact, or impersonate an individual.
123	Theft.Other	Any other incident involving the intentional stealing of physical property or digital data not covered by specific theft subcategories.
124	Other.Uncategorised	Any incident that does not fit into one of the predefined categories in this taxonomy.
125	Other.Undetermined	An incident whose category is currently unknown, under investigation, or cannot be determined.
126	Other.Test	An incident generated solely for the purpose of testing systems, processes, or training personnel.
127	Other.ext-value	A value used to indicate that this attribute is extended and the actual value is provided using the

uank	Keyword	Description
		corresponding ext-* attribute. (see :ref:private_extension_enum)

Table 7: enum_alert_category

C.3. Analyzer Category

Rank	Keyword	Description
0	Undetermined	Analyser category is undetermined
1	APP.BAST	Bastion Host - Secure remote access gateway
2	APP.DAST	Dynamic Application Security Testing - Runtime application testing
3	APP.IAST	Interactive Application Security Testing - Hybrid application testing
4	APP.RASP	Runtime Application Self-Protection - Self-protecting applications
5	APP.SAST	Static Application Security Testing - Source code analysis
6	CLD.CASB	Cloud Access Security Broker - Cloud service security mediation
7	CLD.CIEM	Cloud Infrastructure Entitlement Management - Cloud permission management
8	CLD.CSPM	Cloud Security Posture Management - Cloud configuration monitoring
9	CLD.CWPP	Cloud Workload Protection Platform - Cloud workload security
10	DDoS.ANTI-DDOS	Distributed Denial of Service Protection - DDoS mitigation system
11	DDoS.SCRUB	Scrubber/Scrubbing Center - Traffic cleaning for DDoS
12	DDoS.WAF-DDOS	Web Application Firewall with DDoS - Integrated DDoS protection
13	EMAIL.ANTI-PHISH	Anti-Phishing - Phishing attempt detection
14	EMAIL.DMARC	Domain-based Message Authentication - Email authentication monitoring
15	EMAIL.SEG	Secure Email Gateway - Comprehensive email security
16	EMAIL.SPAM-FILTER	Spam Filter - Unsolicited email detection

Rank	Keyword	Description
17	END.AM	Application Allowlisting - Application execution control
18	END.AV	Antivirus - Signature-based malware detection
19	END.DLP-EP	Endpoint Data Loss Prevention - Endpoint data leakage prevention
20	END.EDR	Endpoint Detection and Response - Advanced endpoint threat hunting
21	END.EPP	Endpoint Protection Platform - Comprehensive endpoint security
22	END.HIDS	Host Intrusion Detection System - Host-based threat monitoring
23	END.HIPS	Host Intrusion Prevention System - Host-based threat prevention
24	END.HPT	Honeypot - Deception-based threat detection
25	END.RASP	Runtime Application Self-Protection - In-app runtime protection
26	ID.DCAP	Data-Centric Audit and Protection - Data-centric security monitoring
27	ID.DLP	Data Loss Prevention - Data leakage prevention across channels
28	ID.IAM	Identity and Access Management - Identity governance and access control
29	ID.IRM	Identity Risk Management - Identity-based risk analysis
30	ID.PAM	Privileged Access Management - Privileged access management
31	ID.PIM	Privileged Identity Management - Privileged account security
32	ID.UEBA	User and Entity Behavior Analytics - Behavioral threat detection
33	NET.DNS-FW	DNS Firewall - Malicious domain filtering
34	NET.DPI	Deep Packet Inspection - Advanced packet analysis
35	NET.FW	Firewall - Network traffic filtering and policy enforcement
36	NET.NAC	Network Access Control - Endpoint compliance and access enforcement

Rank	Keyword	Description
37	NET.NBAD	Network Behavior Anomaly Detection - Anomaly detection in network behavior
38	NET.NDR	Network Detection and Response - Advanced network threat hunting
39	NET.NGFW	Next-Generation Firewall - Advanced firewall with app awareness
40	NET.NIDS	Network Intrusion Detection System - Network traffic analysis for threats
41	NET.NIPS	Network Intrusion Prevention System - Inline threat prevention
42	NET.PROXY	Proxy Server - ACL and TLS session monitoring
43	NET.WAF	Web Application Firewall - HTTP/HTTPS traffic filtering
44	NET.WIDS	Wireless Intrusion Detection System - WiFi threat detection
45	NET.WIPS	Wireless Intrusion Prevention System - WiFi threat prevention
46	OT.IoT-IDS	IoT Intrusion Detection System - IoT device threat detection
47	OT.OT-IDS	Operational Technology IDS - Industrial control system monitoring
48	OT.PLC-SCAN	PLC Scanner - PLC/SCADA vulnerability detection
49	PHY.1D-LAS	1D Laser Sensor - Basic laser presence/distance detection
50	PHY.1D-LiDAR	1D Light Detection and Ranging Sensor - Single-beam laser for distance measurement
51	PHY.2D-LAS	2D Laser Sensor - Planar laser scanning
52	PHY.2D-LiDAR	2D Light Detection and Ranging Sensor - Planar laser scanning for 2D mapping
53	PHY.3D-LAS	3D Laser Sensor - 3D laser scanning
54	PHY.3D-LiDAR	3D Light Detection and Ranging Sensor - 3D environmental scanning and mapping
55	PHY.ACCESS-CTRL	Access Control System - Physical entry/exit control monitoring
56	PHY.ADS	Anti-Drone System - Drone detection and countermeasure system

Rank	Keyword	Description
57	PHY.FR-CAM	Face Recognition Camera - Biometric facial recognition system
58	PHY.GLASS-BRK	Glass Break Detector - Acoustic glass breakage detection
59	PHY.HAR	Human Activity Recognition - AI-based human behavior and motion analysis
60	PHY.LWIR	Long-Wave Infrared - Long-wave thermal imaging
61	PHY.MOT-SEN	Motion Sensor - PIR/ microwave motion detection
62	PHY.MWIR	Mid-Wave Infrared - Mid-wave thermal imaging
63	PHY.OBJ-DET	Object Detection Camera - General object detection and classification
64	PHY.SWIR	Short-Wave Infrared - Short-wave infrared imaging
65	PHY.VAD	Voice Activity Detection - Audio analysis for voice/speech detection
66	PHY.VNIR	Visible and Near-Infrared - Multi-spectral imaging sensor
67	SIEM.ETL	Extract, Transform, Load - Data pipeline tools (Logstash, Fluentd, Vector)
68	SIEM.LOG	Log Analyzer - Log aggregation and analysis (e.g., ELK Stack, Splunk)
69	SIEM.NMS	Network Management System - Network monitoring and management
70	SIEM.SIEM	Security Information and Event Management - Centralized security logging and alerting
71	SIEM.SOAR	Security Orchestration and Response - Automated incident response
72	TI.CTI	Cyber Threat Intelligence - Strategic threat intelligence
73	TI.TI-FEED	Threat Intelligence Feed - External threat data streams
74	TI.TIP	Threat Intelligence Platform - Threat data aggregation and analysis
75	VM.ASM	Attack Surface Management - External attack surface monitoring
76	VM.PENTEST	Penetration Testing Tools - Manual/automated security testing

Rank	Keyword	Description
77	VM.VULN-SCANNER	Vulnerability Scanner - Automated vulnerability assessment
78	ext-value	A value used to indicate that this attribute is extended and the actual value is provided using the corresponding ext-* attribute. (see Section 4.1.1)

Table 8: Analyzer Categories

C.4. Analyzer Data

Rank	Item	Description
0	Undetermined	The analyzed data type is unknown or not yet classified. Used as a placeholder when no other value applies. Example: a generic analyzer whose input could not be identified.
1	Acoustics	Sound pressure or specific frequency measurements captured by a microphone or acoustic sensor. Example: gunshot detection, glass break detection, machine noise spectrum.
2	Alert	IDMEFv2 or equivalent alert produced by an upstream analyzer and consumed as input by a correlator, aggregator or hypervisor. Example: IDMEFv2 JSON alert from a local SOC processed by a central SOC correlator.
3	Contact	Physical interaction or binary state change detected by a contact sensor. Example: door contact switch, cabinet open detection, tamper switch.
4	Content	Payload or body of a communication inspected at the application layer. Example: HTTP body, email message, SMTP DATA section, file transfer payload, DPI-matched bytes.
5	Data	Generic or opaque data stream whose structure is not further specified. Example: raw sensor telemetry, vendor-specific binary stream, serial output.

Rank	Item	Description
6	Datagram	Raw packet-level data captured at the network layer. Example: Ethernet frame, IPv4/IPv6 packet, TCP segment, UDP datagram, ICMP message.
7	External	Data originating from a third party such as a threat intelligence feed or an external CERT. Example: STIX/TAXII feed, IOC list, external CERT notification, vendor advisory.
8	File	Content or metadata of a file read by the analyzer for inspection, integrity or change detection. Example: file hash, PE header, document content, file permissions, modification timestamp.
9	Flow	Network flow records aggregating packets by 5-tuple or similar criteria. Example: NetFlow, IPFIX, sFlow records exported by routers or probes.
10	Fog	Visibility reduction or optical density measured by a fog sensor. Example: fog detection on a perimeter road, visibility monitoring for cameras.
11	Humidity	Relative humidity or moisture content in the air measured by a hygrometer. Example: datacenter humidity level, warehouse moisture monitoring.
12	Images	Visible-spectrum image or video frames captured by a camera. Example: CCTV frame, IP camera stream, facial recognition input.
13	Lidar	Point cloud or distance measurements produced by a laser-based sensor. Example: 3D point cloud from a rotating lidar, perimeter scanning lidar.
14	Light	Ambient light level or flicker detected by a photodetector. Example: lux measurement, room lighting state, strobe or flicker detection.
15	Log	Structured or unstructured event records produced by systems, applications or security devices. Example: syslog messages,

Rank	Item	Description
		Windows EventLog entries, journald records, firewall logs, application traces.
16	Magnetic	Magnetic field strength or anomaly detected by a magnetometer. Example: door open/close magnetic contact, vehicle detection, magnetic anomaly near equipment.
17	Particles	Concentration of dust, smoke or airborne contaminants measured by a particle sensor. Example: smoke detector, PM2.5 sensor, dust monitoring in a facility.
18	PCAP	Packet capture files stored on disk and analyzed offline. Example: .pcap and .pcapng files produced by tcpdump, dumpcap or Wireshark.
19	Rain	Precipitation detected or measured by a rain sensor. Example: rainfall detection on an outdoor site, rain gauge.
20	Seismic	Ground motion or earthquake-related signals captured by a seismograph. Example: earthquake detection, ground vibration near a datacenter.
21	SNMP	Data retrieved via the Simple Network Management Protocol from managed devices. Example: SNMP GET/WALK results, MIB object values, traps and informs.
22	Temperature	Ambient, device or surface temperature measured by a thermometer or thermal probe. Example: server room temperature, device CPU temperature, cold-chain monitoring.
23	Thermal	Infrared image or heat signature data captured by a thermal sensor. Example: thermal camera frame, heat map of a room or equipment.
24	Vibration	Mechanical oscillation or structural movement measured by an accelerometer or vibration sensor. Example: seismic-grade vibration of a

Rank	Item	Description
		server rack, structural health monitoring of a bridge.
25	Water	Presence, flow or level of water detected by a sensor. Example: leak detection under a raised floor, flood sensor, water flow meter.
26	ext-value	A value used to indicate that this attribute is extended and the actual value is provided using the corresponding ext-* attribute. (see Section 4.1.1)

Table 9: enum_analyzer_data

C.5. Analyzer Method

Rank	Keyword	Description
0	Undetermined	Analyser method is undetermed
1	AI	An analyzer that uses machine learning, deep learning, or other artificial intelligence techniques to learn normal behavior and detect sophisticated or novel threats.
2	Anomaly	An analyzer that identifies deviations from established norms or baselines without relying on predefined signatures, often flagging unusual patterns in traffic, behavior, or system activity.
3	Behavioral	An analyzer that monitors and analyzes the actions of users, entities, or processes over time to detect malicious or suspicious activities that deviate from expected behavior patterns.
4	Biometric	An analyzer that uses electronic devices to capture and measure unique physical or behavioral characteristics (e.g., fingerprint, iris, voice) for identification or authentication purposes.
5	Blackhole	A method that analyses traffic destined for a non-existent or sinkhole route to identify malicious activity, such as connections to known command-and-control servers or scanning from infected hosts.

Rank	Keyword	Description
6	Contextual	An analyzer that enriches raw events with additional context (e.g., asset value, user role, time of day, business criticality) to prioritize alerts and assess true impact.
7	Correlation	An analyzer that aggregates and examines multiple, disparate data streams or events to identify complex relationships, patterns, sequences, or dependencies that indicate a security incident.
8	Ensemble	An analyzer that combines multiple detection methods (e.g., signature, anomaly, behavioral) to improve accuracy, reduce false positives, and detect threats that single methods might miss.
9	Fingerprinting	An analyzer that creates unique identifiers or "fingerprints" for devices, applications, or network stacks to detect spoofing, unauthorized devices, or configuration changes.
10	Frequency	An analyzer that detects incidents based on the rate or regularity of events (e.g., repeated failed logins, rapid-fire requests) exceeding or falling below expected frequencies.
11	Fusion	An analyzer that combines data from multiple heterogeneous sensors and sources to create a comprehensive, high-confidence view of an incident, reducing ambiguity and false alerts.
12	Geolocation	An analyzer that determines the physical location of an asset or event (e.g., login attempt, IP address, device) and flags activities occurring from unexpected or high-risk locations.
13	Graph-based	An analyzer that models relationships between entities (users, devices, processes) as graphs and detects anomalies or attack paths by analyzing connections, dependencies, and traversals.

Rank	Keyword	Description
14	Heat	An analyzer (sensor or device) that detects, measures, and monitors thermal energy (infrared radiation) to identify anomalies like fires, overheating equipment, or human presence.
15	Heuristic	An analyzer that detects potentially unknown threats by using algorithmic logic, rules of thumb, or suspicious characteristics rather than relying on specific signature matches.
16	Honeypot	A decoy system or resource designed to lure, detect, and analyze malicious activity by mimicking a legitimate target, diverting attackers away from real assets.
17	Hygiene	An analyzer that continuously checks systems and configurations against security best practices, compliance standards, or hardening guidelines to identify weaknesses or drift.
18	Integrity	An analyzer that monitors critical system components (files, configurations, registry keys) for unauthorized changes, verifying their integrity against a known good baseline.
19	Metadata	An analyzer that examines the data about data (e.g., file creation timestamps, email headers, connection logs) to uncover hidden relationships or suspicious attributes.
20	Monitor	An analyzer that continuously observes a system, network, or environment to track its state, health, or activity, often providing real-time alerts on specific conditions.
21	Movement	An analyzer (sensor or system) that detects, tracks, and quantifies physical motion using technologies like radar, lidar, or video analytics.
22	Orchestration	An analyzer that coordinates and triggers automated response actions based on detected

Rank	Keyword	Description
		incidents, often integrated with SOAR (Security Orchestration, Automation, and Response) platforms.
23	Pattern	An analyzer that identifies specific sequences, combinations, or recurring arrangements of events or data that indicate malicious activity, even if individual elements appear benign.
24	Policy	An analyzer that evaluates events, configurations, or behaviors against a set of predefined rules, configurations, or compliance requirements to detect violations or misconfigurations.
25	Predictive	An analyzer that uses historical data and modeling to forecast potential future incidents, vulnerabilities, or attack vectors before they occur.
26	Protocol	An analyzer that validates network traffic or communications against expected protocol specifications, RFC compliance, or standard behavior to detect anomalies or malicious variations.
27	Recon	An analyzer that actively or passively probes or monitors an environment to discover assets, services, or vulnerabilities, often as part of a defensive assessment or adversary simulation.
28	Reputation	An analyzer that evaluates the trustworthiness of an entity (e.g., IP address, domain, file hash) by checking it against known threat intelligence lists, blocklists, or reputation scores.
29	Rule-based	An analyzer that applies conditional logic (if-then-else statements) defined by experts to correlate events and generate alerts based on specific combinations of conditions.
30	Sequence	An analyzer that detects threats by examining the order and

Rank	Keyword	Description
		timing of events, identifying attack chains or kill chain progressions (e.g., scan # exploit # installation # C2).
31	Signature	An analyzer that detects known threats by matching events or patterns against a database of specific signatures, hashes, or Indicators of Compromise (IoCs).
32	Statistical	An analyzer that detects anomalies by establishing a baseline of normal behavior and identifying events that deviate significantly from expected statistical parameters.
33	Tarpit	A mechanism that intentionally slows down or delays suspicious connections (e.g., network connections or login attempts) to hinder automated attacks and scanning.
34	Threat Intelligence	An analyzer that ingests and matches internal events against external threat feeds, IoC lists, and adversary TTPs (Tactics, Techniques, and Procedures) to identify known threats.
35	Threshold	An analyzer that detects incidents by comparing a metric or count (e.g., number of failed logins, traffic volume) against a predefined limit or threshold.
36	Trend	An analyzer that monitors data over extended periods to identify gradual changes, emerging patterns, or long-term shifts that may indicate evolving threats or security degradation.
37	ext-value	A value used to indicate that this attribute is extended and the actual value is provided using the corresponding ext-* attribute. (see Section 4.1.1)

Table 10: Analyzer Methods

C.6. Source Category

Rank	Keyword	Description
0	Undetermined	Source Category is undetermined.

Rank	Keyword	Description
1	Acoustic.Sound.GlassBreak	Specific frequencies (glass breaking, gunshots)
2	Acoustic.Sound.Infrasound	Infrasound (nuclear explosions, volcanic activity)
3	Acoustic.Sound.PressurizedLeak	Leaks in pressurized pipes (high-frequency hiss)
4	Acoustic.Sound.SonarEcho	Sonar echoes (submarines, fish, seafloor mapping)
5	Acoustic.Sound.UltrasonicNoise	Ultrasonic noise (for pest detection or leak detection)
6	Acoustic.Vibration.Footstep	Footsteps (geophones or seismic sensors)
7	Acoustic.Vibration.Seismic	Seismic activity (earthquakes, tremors)
8	Acoustic.Vibration.StructuralCrack	Structural integrity (cracks via acoustic emission)
9	Acoustic.Vibration.VehicleVibe	Vehicle movement (magnetic or seismic)
10	Acoustic.Vibration.VibAnalysis	Machinery imbalance (vibration analysis)
11	ChemBio.Gases.CO	Carbon Monoxide (CO)
12	ChemBio.Gases.CWA	CWA (Chemical Warfare Agents - Nerve gas, Blister agents)
13	ChemBio.Gases.Freon	Refrigerant leaks (Freon)
14	ChemBio.Gases.Hydrogen	Hydrogen
15	ChemBio.Gases.Methane	Natural Gas / Methane
16	ChemBio.Gases.O2Level	Oxygen levels (deficiency or enrichment)
17	ChemBio.Gases.Propane	Propane / LPG
18	ChemBio.Gases.Radon	Radon
19	ChemBio.Gases.VOC	Volatile Organic Compounds (VOCs) - paint fumes, off-gassing
20	ChemBio.Properties.ORP	Oxidation-Reduction Potential (ORP)
21	ChemBio.Properties.pHLevel	pH levels (acidity/alkalinity)
22	ChemBio.Properties.Salinity	Salinity
23	ChemBio.Threats.Anthrax	Anthrax spores
24	ChemBio.Threats.Botulinum	Botulinum toxin
25	ChemBio.Threats.Pathogen	Pathogens in water (E. coli, Cholera)
26	ChemBio.Threats.Ricin	Ricin
27	Cyber.Application.SQLQuery	Malicious code injected into a database query (SQL Injection).
28	Cyber.Application.XSSPayload	The script injected into a webpage to steal data or redirect users.
29	Cyber.Cloud.APIToken	A compromised key used to access cloud resources (AWS, Azure) or SaaS platforms.

Rank	Keyword	Description
30	Cyber.Cloud.BucketName	The specific cloud storage location where data was stored or exfiltrated from.
31	Cyber.Cloud.CloudResourceID	The unique ID of the compromised virtual machine or database.
32	Cyber.Cloud.OAuthApp	A malicious third-party app granted permissions to a user's data.
33	Cyber.Email.EmailAuthResult	The authentication result showing if the email passed security checks.
34	Cyber.Email.MessageID	A unique identifier for the email used to track it across mail servers.
35	Cyber.Email.RecipientEmail	The target email address of the attack.
36	Cyber.Email.ReplyToEmail	The address where replies go; often different from the sender for tracking.
37	Cyber.Email.ReturnPath	The address where non-delivery receipts go; used in spoofing analysis.
38	Cyber.Email.SenderEmail	The "From" address used in a phishing campaign.
39	Cyber.Email.XOrigIP	The original IP address of the sender (if recorded by the mail server).
40	Cyber.Endpoint.CmdLine	The exact command line used to execute the malware; often reveals attacker intent.
41	Cyber.Endpoint.DLL	The name of a malicious library loaded into a legitimate process (DLL Sideloading).
42	Cyber.Endpoint.FileHash	The cryptographic fingerprint of the malicious file (SHA256, MD5).
43	Cyber.Endpoint.FilePath	The location on the file system where the malware resides.
44	Cyber.Endpoint.MemoryArtifact	Code or processes that exist only in RAM (volatile memory) and not on disk.
45	Cyber.Endpoint.PID	The unique identifier of the malicious process at the time of analysis.
46	Cyber.Endpoint.ProcessName	The name of the malicious executable file running on the endpoint.
47	Cyber.Endpoint.RegistryKey	The Windows Registry path used for persistence or configuration changes.

Rank	Keyword	Description
48	Cyber.Endpoint.ScheduledTask	The name of a task created by the attacker for persistence or execution.
49	Cyber.Endpoint.ServiceName	The name of a malicious service installed to run the malware.
50	Cyber.Forensics.NetFlow	Metadata about the connection (who talked to whom, when, and data volume).
51	Cyber.Forensics.PacketPayload	The raw data being transmitted containing passwords or exfiltrated data.
52	Cyber.Hardware.DeviceSerial	The unique identifier of a specific compromised laptop or mobile device.
53	Cyber.Hardware.USBDeviceID	The specific make/model/ serial of a USB drive used to introduce malware.
54	Cyber.Identity.LoginTimestamp	The specific time of an unusual login (e.g., 3:00 AM from a foreign location).
55	Cyber.Identity.Username	The specific user account that was compromised or used for lateral movement.
56	Cyber.Network.IPv4	The source IP of an attacker, a Command & Control (C2) server, or a malicious scan.
57	Cyber.Network.IPv6	The source IP in IPv6 format; increasingly used to bypass IPv4-based allowlists.
58	Cyber.Network.MAC	The physical device identifier on a local network; useful if the attacker is on-site.
59	Cyber.Network.Port	The specific source or destination port used (e.g., 445 for ransomware spreading).
60	Cyber.Network.Protocol	The network protocol used; anomalous use may indicate tunneling (e.g., DNS tunneling).
61	Cyber.Web.Domain	The domain hosting malware or Command & Control infrastructure (e.g., evil.com).
62	Cyber.Web.HTTPMethod	Unusual HTTP methods like PUT or DELETE used to manipulate data.
63	Cyber.Web.ReferrerHeader	The page the user was on before clicking the malicious link.
64	Cyber.Web.SessionID	A hijacked session token used to impersonate a user without a password.
65	Cyber.Web.URIPath	The specific directory or file requested on the web server.

Rank	Keyword	Description
66	Cyber.Web.URL	The full malicious link pointing to a payload or phishing page.
67	Cyber.Web.UserAgent	The browser or application signature used by the attacker when connecting.
68	ElectroMag.Infrared.GasLeakIR	Gas leaks (specific gases absorb specific IR wavelengths)
69	ElectroMag.Infrared.HeatTemp	Heat (temperature of objects)
70	ElectroMag.Infrared.MoistureContent	Moisture content (in materials)
71	ElectroMag.Infrared.ThermalGradient	Thermal gradients (heat loss in buildings)
72	ElectroMag.Infrared.WarmBody	Warm bodies (humans, animals) via Passive Infrared (PIR)
73	ElectroMag.Ultraviolet.BioContamUV	Biological contamination
74	ElectroMag.Ultraviolet.CoronaArc	Corona discharge (electrical arcing)
75	ElectroMag.Ultraviolet.FlameUV	Flames (fire detection)
76	ElectroMag.Ultraviolet.ForgeryUV	Forgery (detecting UV security features on documents)
77	ElectroMag.Light.BiometricEye	Biological signatures (retinas, irises) - Failed recognition indicates intrusion attempt
78	ElectroMag.Light.EdgesShapes	Edges and shapes (for machine vision)
79	ElectroMag.Light.Flicker	Flicker (from faulty lighting or machinery)
80	ElectroMag.Light.LightColor	Color and wavelength (spectral content)
81	ElectroMag.Light.LightPresence	Presence of light (ambient brightness)
82	ElectroMag.Light.MotionOptical	Motion (optical flow, changes in pixel arrays)
83	ElectroMag.Light.ObjectsGeneric	Specific objects (via computer vision: faces, vehicles, text, barcodes)
84	Force.Atmospheric.Altitude	Altitude (in aircraft)
85	Force.Atmospheric.BaroPress	Barometric pressure (weather changes)
86	Force.Atmospheric.VacuumLevel	Vacuum levels (in manufacturing)
87	Force.Fluid.Airflow	Airflow (differential pressure)
88	Force.Fluid.WaterPressure	Water pressure (for leak detection via pressure drop)
89	Force.Touch.GripStrength	Grip strength
90	Force.Touch.PhysicalContact	Physical contact (bumpers on robots, touch screens)
91	Force.Touch.TirePressure	Tire pressure
92	Force.Touch.Weight	Weight (scales, load cells)
93	Radiation.General.Alpha	Alpha particles
94	Radiation.General.Beta	Beta particles
95	Radiation.General.Cosmic	Cosmic rays
96	Radiation.General.Gamma	Gamma rays

Rank	Keyword	Description
97	Radiation.General.Neutron	Neutron radiation
98	Radiation.General.RadonDecay	Radon gas (decay products)
99	Radiation.General.Xray	X-rays
100	Magnetic.General.CurrentFlow	Current flow (Hall effect sensors, clamp meters)
101	Magnetic.General.ESD	Electrostatic discharge
102	Magnetic.General.Ferrous	Ferrous metals (via magnetic disturbance)
103	Magnetic.General.MagneticAnomaly	Magnetic anomalies (submarines, buried structures)
104	Magnetic.General.Orientation	Orientation (Compass / Magnetometer - detecting Earth's magnetic field)
105	Magnetic.General.Powerline	Power lines (AC magnetic fields)
106	Magnetic.General.Voltage	Voltage (non-contact voltage testers)
107	Object.Indoor.Container	Bottles / Cups / Cutlery
108	Object.Indoor.Document	Books / Papers / Documents
109	Object.Indoor.Electronics	Electronics (laptops / phones / TVs)
110	Object.Indoor.Furniture	Furniture (chairs / tables / sofas)
111	Object.Indoor.Weapon	Weapons (guns / knives)
112	Object.LivingBeings.Animal	Animals (general wildlife/ livestock/pets)
113	Object.LivingBeings.BodyPose	Pose (skeleton keypoints / body position)
114	Object.LivingBeings.EyeGaze	Eyes (gaze tracking / eye blink)
115	Object.LivingBeings.Face	Faces (facial recognition)
116	Object.LivingBeings.HandGesture	Hands (hand gestures / raising hand)
117	Object.LivingBeings.Human	Humans (person/people detection)
118	Object.Outdoor.Building	Buildings / Houses
119	Object.Outdoor.Pothole	Potholes / Road cracks
120	Object.Outdoor.RoadSign	Road signs / Traffic lights
121	Object.Outdoor.Vegetation	Trees / Plants / Vegetation
122	Object.SpecificScenes.FireVisual	Fire / Flames (visual spectrum)
123	Object.SpecificScenes.Flood	Flooding / Water accumulation
124	Object.SpecificScenes.SmokeVisual	Smoke plume (visual spectrum)
125	Object.SpecificScenes.Snow	Snow coverage
126	Object.Vehicles.Aircraft	Aircraft (drones / planes / helicopters)
127	Object.Vehicles.Boat	Boats / Ships
128	Object.Vehicles.Bus	Buses
129	Object.Vehicles.Car	Cars / Automobiles
130	Object.Vehicles.LicensePlate	License plates (Automatic Number Plate Recognition - ANPR)
131	Object.Vehicles.Moto	Motorcycles / Bicycles
132	Object.Vehicles.Truck	Trucks

Rank	Keyword	Description
133	Particulate.Aerosols.Bioaerosol	Viral or bacterial particles (bio-aerosol detectors)
134	Particulate.Aerosols.Dust	Airborne dust (particulate counters)
135	Particulate.Aerosols.Pollen	Pollen and allergens
136	Particulate.Aerosols.SmokeIonization	Combustion particles (Ionization smoke detectors)
137	Particulate.Aerosols.SmokePhotoelectric	Visible smoke obscuration (Photoelectric smoke detectors)
138	Particulate.Liquids.ChemicalSpill	Chemical spills (specific conductivity or pH changes)
139	Particulate.Liquids.FuelSheen	Fuel/Oil sheen on water
140	Particulate.Liquids.LiquidLevel	Level of liquid in a tank (ultrasonic, capacitive, float)
141	Particulate.Liquids.WaterLeak	Water leakage (conductivity or float switches)
142	Particulate.Solids.DensityChange	Density changes (in manufacturing)
143	Particulate.Solids.ForeignObject	Foreign objects in food (X-ray machines)
144	Particulate.Solids.GrainFlow	Grain flow (in agriculture)
145	Particulate.Solids.Landmine	Landmines/UXO (Ground Penetrating Radar)
146	Particulate.Solids.Metal	Metal (Metal detectors via eddy currents)
147	Software.Detected.Crowd	Crowd formation
148	Software.Detected.CyberAttack	Cyber-attack (Anomalous network traffic patterns)
149	Software.Detected.Drowsiness	Drowsiness (Eye closure detection via camera)
150	Software.Detected.Explosion	Explosion (Sudden light flash + pressure wave + sound)
151	Software.Detected.FailedBiometric	Failed retina/iris scan (multiple failures may indicate intrusion)
152	Software.Detected.FireFusion	Fire (Combination of smoke, heat, and flicker)
153	Software.Detected.Intrusion	Intrusion (Crossing a virtual line)
154	Software.Detected.Loitering	Loitering (Person staying too long)
155	Software.Detected.Tailgating	Tailgating (Following someone through a door)
156	ext-value	A value used to indicate that this attribute is extended and the actual value is provided using the corresponding ext-* attribute. (see Section 4.1.1)

Table 11: Source Categories

C.7. Target Category

Rank	Keyword	Description
0	Undetermined	Target Category is undetermined
1	Commercial.ClientBase	Client portfolio, active contracts
2	Commercial.Partnerships	Strategic alliances, partner networks
3	Commercial.Reputation	Brand image, online reputation, goodwill
4	Financial.Cash	Liquidities, bank accounts
5	Financial.Investments	Shares, bonds, holdings in subsidiaries
6	Financial.Receivables	Customer receivables, promissory notes
7	Furniture.Fittings	Lighting, partitions, flooring
8	Furniture.HVAC	Air conditioning, heating systems (non-industrial)
9	Furniture.OfficeFurniture	Desks, chairs, filing cabinets
10	Human.EmployerBrand	Employment contracts, recruitment appeal
11	Human.Organization	Work processes, internal know-how
12	Human.Skills	Employee expertise, team competencies
13	Infrastructure.CivilWorks	Bridges, loading docks, quays
14	Infrastructure.Foundations	Special foundations, internal roads, pavements
15	Infrastructure.Networks	Buried water, gas, electricity, fiber networks
16	IP.Copyrights	Software code, manuals, artistic works
17	IP.Patents	Registered patents, inventions
18	IP.Trademarks	Brand names, logos, registered designs
19	IT.Endpoints	Computers, printers, scanners
20	IT.Hardware	Servers, storage arrays, cabling
21	IT.Telecom	WiFi access points, phones, smartphones
22	Logistics.Containers	Containers, trailers, wagons
23	Logistics.SpecialVehicles	Mobile cranes, aerial work platforms
24	Logistics.Vehicles	Cars, trucks, forklifts
25	Marketing.Content	Market studies, ad campaigns, social media
26	Marketing.Domains	Website URLs, domain names
27	Marketing.Identity	Visual identity, logo, brand guidelines
28	Production.HeavyEquipment	Presses, furnaces, compressors, generators
29	Production.Machinery	Assembly lines, industrial robots, conveyors

Rank	Keyword	Description
30	Production.MachineTools	Lathes, milling machines, industrial 3D printers
31	RealEstate.Buildings	Factories, warehouses, offices, commercial premises
32	RealEstate.Land	Unbuilt land, forests, or agricultural grounds
33	RealEstate.SpecialStructures	Hangars, silos, parking lots
34	Software.Applications	ERP, CRM, SaaS platforms
35	Software.Data	Customer databases, supplier lists, prospects
36	Software.Licenses	Software usage rights, subscriptions
37	ext-value	A value used to indicate that this attribute is extended and the actual value is provided using the corresponding ext-* attribute. (see Section 4.1.1)

Table 12: Target Categories

Appendix D. Examples

This section contains several examples of events/incidents which may be described using the IDMEF Data Model defined in.

For each example, the serialization method listed in Section 5 was used on the original IDMEF message to produce a JSON representation.

D.1. Physical intrusion

Listing 1 describes an incident where an unidentified man was detected on company premises near the building where server room A is located.

```
{
  "Version": "2.D.V07",
  "ID": "819df7bc-35ef-40d8-bbee-1901117370b1",
  "Description": "Potential intruder detected",
  "Type": "Physical",
  "Priority": "Low",
  "Status": "Incident",
  "Cause": "Malicious",
  "CreateTime": "2021-05-10T16:52:13.075994+00:00",
  "StartTime": "2021-05-10T16:52:13+00:00",
  "Category": [
    "Access.Forced"
  ],
  "Analyzer": {
    "Name": "BigBrother",
    "Hostname": "bb.acme.com",
    "Model": "Big Brother v42",
    "Category": [
      "PHY.HAR",
      "FRC.FR-CAM"
    ],
    "Data": [
      "Images"
    ],
    "Method": [
      "Movement",
      "Biometric",
      "AI"
    ],
    "IP": "192.0.2.1"
  },
  "Sensor": [
    {
      "IP": "192.0.2.2",
      "Name": "Camera #23",
      "Model": "SuperDuper Camera v1",
      "Location": "Hallway to server room A1"
    }
  ],
  "Source": [
    {
      "Note": "Black Organization, aka. APT 4869"
    }
  ],
  "Attachment": [
    {
      "Name": "wanted",
      "FileName": "fbi-wanted-poster.jpg",
      "Size": 1234567,
      "Ref": ["https://www.fbi.gov/wanted/topten"],
      "ContentType": "image/jpg",
      "ContentEncoding": "base64",
      "Content": "..."
    },
    {
      "Name": "pic01",
      "Note": "Hi-res picture showing John Doe near server room A1",
      "ExternalURI": ["ftps://192.0.2.1/cam23/20210510165211.jpg"],
      "ContentType": "image/jpg"
    }
```

D.2. Cyberattack

Listing 2 describes an incident related to a potential bruteforce attack against the "root" user account of the server at 192.0.2.2 and 2001:db8::/32.

```
{
  "Version": "2.D.V07",
  "ID": "819df7bc-35ef-40d8-bbee-1901117370b2",
  "Description": "Potential bruteforce attack on root user account",
  "Type": "Cyber",
  "Priority": "Medium",
  "CreateTime": "2021-05-10T16:55:29.196408+00:00",
  "StartTime": "2021-05-10T16:55:29+00:00",
  "Category": [
    "Access.Forced"
  ],
  "Analyzer": {
    "Name": "SIEM",
    "Hostname": "siem.acme.com",
    "Model": "Concerto SIEM 5.2",
    "Category": [
      "SIEM.SIEM",
      "SIEM.LOG"
    ],
    "Data": [
      "Log"
    ],
    "Method": [
      "Monitor",
      "Signature"
    ],
    "IP": "192.0.2.1"
  },
  "Sensor": [
    {
      "IP": "192.0.2.5",
      "Name": "syslog",
      "Hostname": "www.acme.com",
      "Model": "rsyslog 8.2110",
      "Location": "Server room A1, rack 10"
    }
  ],
  "Target": [
    {
      "IP": "192.0.2.2",
      "Hostname": "www.acme.com",
      "Location": "Server room A1, rack 10",
      "User": "root"
    },
    {
      "IP": "2001:db8::/32",
      "Hostname": "www.acme.com",
      "Location": "Server room A1, rack 10",
      "User": "root"
    }
  ]
}
```

D.3. Server outage

Listing 3 describes an incident where the webserver at "www.example.com" encountered some kind of failure condition resulting in an outage.

```
{
  "Version": "2.D.V07",
  "ID": "819df7bc-35ef-40d8-bbee-1901117370b3",
  "Description": "A server did not reply to an ICMP ping request",
  "Type": "Availability",
  "Priority": "Medium",
  "Status": "Incident",
  "Cause": "Unknown",
  "CreateTime": "2021-05-10T16:59:11.875209+00:00",
  "StartTime": "2021-05-10T16:59:11.875209+00:00",
  "Category": [
    "Availability.Outage"
  ],
  "Analyzer": {
    "Name": "NMS",
    "Hostname": "nms.example.com",
    "Model": "Concerto NMS 5.2",
    "Category": [
      "SIEM.NMS"
    ],
    "Data": [
      "Network"
    ],
    "Method": [
      "Monitor"
    ],
    "IP": "192.0.2.1"
  },
  "Target": [
    {
      "IP": "192.168.1.2",
      "Hostname": "www.acme.com",
      "Service": "website",
      "Location": "Server room A1, rack 10"
    }
  ]
}
```

D.4. Combined incident

Listing 4 describes a combined incident resulting from the correlation of the previous physical, cyber and availability incidents.

```
{
  "Version": "2.D.V07",
  "ID": "819df7bc-35ef-40d8-bbee-1901117370b4",
  "Description": "Intrusion and Sabotage detected",
  "Type": "Combined",
  "Priority": "High",
  "Status": "Incident",
  "Cause": "Malicious",
  "CreateTime": "2021-05-10T16:59:15.075994+00:00",
  "StartTime": "2021-05-10T16:52:11+00:00",
  "Category": [
    "Access.Unauthorized",
    "Sabotage.Data",
  ,
  "CorrelID": [
    "819df7bc-35ef-40d8-bbee-1901117370b1",
    "819df7bc-35ef-40d8-bbee-1901117370b2",
    "819df7bc-35ef-40d8-bbee-1901117370b3"
  ],
  "Analyzer": {
    "Name": "Correlator",
    "Hostname": "correlator.acme.com",
    "Model": "Concerto Hybrid Correlator v5.2",
    "Category": [
    ],
    "Data": [
      "Alert"
    ],
    "Method": [
      "Correlation"
    ],
    "IP": "192.0.2.1"
  },
  "Source": [
    {
      "Note": "Black Organization, aka. APT 4869"
    }
  ],
  "Target": [
    {
      "Location": "Server room A1"
    },
    {
      "IP": "192.0.2.2",
      "Hostname": "www.acme.com",
      "User": "root"
    },
    {
      "IP": "192.0.2.2",
      "Hostname": "www.acme.com",
      "Service": "website"
    }
  ]
}
```

D.5. Hazard incident

Listing 5 describes a heavy snow storm announced in 48h on Paris and Versailles.

```
{
  "Version": "2.D.V07",
  "ID": "819df7bc-35ef-40d8-bbee-1901117370b1",
  "Description": "Snow storm forecast",
  "Priority": "Low",
  "Status": "Incident",
  "Cause": "Hazard",
  "Confidence": 0.8,
  "CreateTime": "2021-05-10T16:52:13.075994+00:00",
  "StartTime": "2021-05-12T10:00:00+00:00",
  "EndTime": "2021-05-12T10:00:00+00:00",
  "Type": ["Physical"],
  "Category": [
    "Meteo.Snow",
    "Meteo.Wind",
    "Meteo.Cold"
  ],
  "Analyzer": {
    "Name": "Weather Monitor",
    "Hostname": "weather.acme.com",
    "IP": "192.0.2.1"
  },
  "Source": [
    {
      "Note": "Heavy snow storm coming from North"
    }
  ],
  "Target": [
    {
      "GeoLocation": "48.8584,2.2945",
      "UnLocation": "FR PAR",
      "Location": "Acme Paris Site"
    },
    {
      "GeoLocation": "48.8019,2.1301",
      "UnLocation": "FR VER",
      "Location": "Acme Versailles Site"
    }
  ]
}
```

Appendix E. JSON Validation Schema (Non-normative)

Listing 5 contains a JSON Schema that can be used to validate incoming IDMEF messages prior to processing. Please note that extraneous linebreaks have been included due to formatting constraints.

```
{
    "description": "JSON schema for the Intrusion Detection Message
 Exchange Format (IDMEF) version 2 (revision 2.D.V08)",
    "properties": {
        "Version": {
            "description": "The version of the IDMEF format in use by this
 alert. During the drafts tuning period the version is equal to the draft
 version. Therefore it is \"2.D.V0X\" for Draft V0X.",
            "enum": [
                "2.D.V08"
            ]
        },
        "ID": {
            "description": "Unique identifier for the alert.",
            "$ref": "#/definitions/uuidType"
        },
        "OrganisationName": {
            "description": "Corporate/Main Office Organisation Name Useful
 if alerts are sent to a multi-organisation central incident detection
 manager. Example: ACME Corporation",
            "type": "string"
        },
        "OrganisationId": {
            "description": "Corporate/Main Office Organisation ID. Where
 possible official organisation ID manage by national authority. Useful
 if alerts are sent to a multi-organisation central incident detection
 manager. This ID has to be chosen depending on the overall detection
 perimeter and the nature of the monitored organisation (Private/Public,
 Commercial, International, etc.) Examples: OrganisationId in France could
 be SIREN, in England could be CR, Germany could be Handelsregisternummer,
 Spain could be CIF, Italia could be Partita IVA, USA could be EIN, etc.
 Commercial OrganisationId in Europe could be V.A.T ID",
            "type": "string"
        },
        "EntityName": {
            "description": "Entity Name, monitored by the organisation,
 where the incident occurred. Could be a town, region or country name
 or an internal name. Could also be the name of a client for a MSSP
 centralizing it's client incidents in a single system. Do not repeat
 the organisation name in the EntityName Example: - ACME HeadQuaters
 is located in Paris France and has a local office in India - If the
 incident occurred in the local office: \"OrganisationName\": \"ACME\" ,
 \"EntityName\": \"India\" - If the incident occurred in the headquaters:
 \"OrganisationName\": \"ACME\", \"EntityName\": \"Headquaters\" (or
 \"Paris\")",
            "type": "string"
        },
        "EntityId": {
            "description": "Entity ID, monitored by the organisation,
 where the incident occurred. Useful if organisation and entity are not
 directly linked, like a client and a MSSP.",
            "type": "string"
        },
        "EntitySector": {
            "description": "The economic sector(s) and sub-sector(s)
 in which the entity operates. Values follow the dot notation
 sector.subsector based on the critical infrastructure taxonomy defined
 in the NIS2 Directive and CER (Critical Entities Resilience) Directive.
 This attribute enables sector-based correlation, regulatory compliance
 reporting, and risk context for incident detection.",
```

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